

Risk reduction of cognitive decline and dementia

WHO guidelines, second edition



Web Annex E

Evidence profile and evidence-to-decision table for management of diabetes



World Health
Organization

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Management of diabetes

Scoping question: For adults with normal cognition or mild cognitive impairment (MCI) and type 2 diabetes mellitus (T2DM), is treatment of T2DM more effective than placebo/no intervention in reducing the risk of cognitive decline and/or dementia?

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Abbreviations

3MSE	modified Mini-Mental State Examination
ADL	activities of daily living
AMSTAR-II	A MeaSurement Tool to Assess Systematic Reviews
CARE	Comprehensive Assessment and Referral Evaluation (CARE) Diagnostic Scale
CI	confidence interval
CRoB	Cochrane Risk of Bias
DACD	diabetes associated cognitive decline
DPP-4i	dipeptidyl peptidase-4 inhibitors
DSST	Digit Symbol Substitution Test
ERT	evidence review team
EtD	Evidence-to-decision
GDG	guideline development group
GLP-1 RA	glucagon-like peptide receptor agonists
GRADE	Grading of Recommendations, Assessment, Development, and Evaluations
HbA1c	glycated haemoglobin
HIC	high-income countries
IADL	instrumental activities of daily living
LMIC	low- and middle-income countries
MCI	mild cognitive impairment
MD	mean difference
MMSE	Mini-Mental State Examination
MoCA	Montreal Cognitive Assessment
NOS	Newcastle-Ottawa Scale
PICO	Population, Intervention, Comparator, Outcome
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
RCTs	randomised controlled trials
RR	risk ratio
SD	standard deviation
SF-36	36-Item Short Form Survey
SMD	standard mean difference
T1DM	type 1 diabetes mellitus
T2DM	type 2 diabetes mellitus
USD	United States dollar
WHO	World Health Organization

1. Background

Dementia is a progressive neurodegenerative condition and a leading cause of disability and dependency among older adults (1). It is estimated to currently affect 57 million people worldwide, with this number projected to triple by 2050 (2). Despite advances in symptomatic management, no cure currently exists, underscoring the importance of finding effective strategies to delay onset or reduce incidence of dementia and mitigate its associated disease burden (1). Key to this is the identification and modification of risk factors for dementia. Among these, diabetes mellitus - a chronic metabolic disorder characterised by persistent hyperglycaemia that results from defects in insulin secretion, insulin action, or both - has emerged as a critical focus due to its growing prevalence (3, 4) and mounting evidence linking it to adverse cognitive outcomes (5, 6).

There are two major types of diabetes. Type 1 Diabetes Mellitus (T1DM) is a disease identified by inadequate insulin production. It is an autoimmune condition of which the root cause is not well understood (3). Conversely, Type 2 Diabetes Mellitus (T2DM) is a condition characterised by insulin resistance or ineffective use of insulin (3), and is strongly associated with lifestyle and metabolic determinants (3). Globally, approximately 830 million people have diabetes with the majority living in low- and middle-income countries (7). Of worldwide diabetes cases, T2DM accounts for over 95% (4).

While both forms of diabetes carry significant health consequences, T2DM has received particular attention in the context of cognitive ageing and dementia risk, given its high prevalence and potential for prevention or control (8). There is a large body of evidence that identifies individuals with T2DM as being at an increased risk of both cognitive decline and dementia (5, 6, 8). The exact mechanisms through which T2DM contributes to this increased risk are unclear, however it is believed to be related to metabolic, vascular, and neurodegenerative changes that compound over time (9). Regardless, reducing the incidence and improving management of T2DM may play a key role in reducing global dementia burden.

A range of strategies are commonly used for the management of T2DM, including both pharmacological (e.g., metformin, sulfonylureas) and lifestyle (e.g., dietary management, physical activity) interventions (10). These interventions have the primary aim of achieving and sustaining glycaemic control, while mitigating disease progression and complications (3, 4, 10). However, while such interventions are well established in the context of T2DM management, it is unclear whether they confer protective effects against cognitive decline/dementia (8, 11).

In 2019, the World Health Organisation (WHO) developed Guidelines on Risk Reduction of Cognitive Decline and Dementia (11). As part of these guidelines and based on systematic evidence derived for T2DM, WHO made the conditional recommendation that **“the management of diabetes may be offered to adults with diabetes to reduce the risk of cognitive decline and/or dementia”** (11). This recommendation was made conditionally, as the quality of the available evidence at that time was very low, and as such uncertainty regarding the benefits of diabetes management on cognitive health remained. In addition, WHO issued a strong

recommendation that **“the management of diabetes in the form of medications and/or lifestyle interventions should be offered to adults with diabetes according to existing WHO guidelines”** based on very low to moderate certainty evidence (11).

Since 2019 the evidence on diabetes management in the context of dementia prevention has continued to amass. As such, the present review was conducted to prepare an updated evidence profile to inform the 2025 WHO Guidelines on Risk Reduction of Cognitive Decline and Dementia. This review aims to provide an up-to-date summary of the available intervention literature examining the effectiveness of the treatment of T2DM (e.g., through medications, or lifestyle interventions) on reducing the risk of cognitive decline and/or dementia. **The review was restricted to T2DM, given mid-life T2DM has most consistently been identified as a risk factor for dementia (8).**

2. Methodology

The procedure for evidence retrieval and synthesis has been guided by Chapter 8 of the WHO Handbook for Guideline Development (12). In accordance with this handbook, the present systematic review was commissioned for the purposes of updating an existing systematic review and to inform the preparation of evidence summaries.

2.1 Question

This systematic review will address the following two questions of interest:

- 1) In people with T2DM, is the treatment of T2DM using glucose-lowering medicines/pharmacological anti-diabetes agents more effective than placebo or no treatment/usual care in lowering the risk of cognitive decline and/or dementia?
- 2) In people with T2DM, is the treatment of T2DM using lifestyle interventions (dietary or physical activity/exercise) more effective than no treatment/usual care in lowering the risk of cognitive decline and/or dementia?

2.2 Eligibility and study selection

Population (P):

- Adults with T2DM who are cognitively normal or who have Mild Cognitive Impairment (MCI)
- Specifically, studies with samples comprising adults with a mean age of 40+ at the first reported measurement or reporting sub-group analysis of adults aged 40+ separately

Intervention (I):

- Glucose-lowering medicines
- Lifestyle Interventions (dietary, or physical activity/exercise)

Comparator (C):

- Usual Care / Placebo / No Intervention

Outcomes (O):

Critical outcomes:

- Dementia
- Incident MCI
- Cognitive function (or cognitive test results using validated measurements)

Important outcomes:

- Quality of Life
- Functional Level (e.g., Activities of Daily Living/ADL, Instrumental Activities of Daily Living/IADL)
- Adverse Events
- Drop-Out Rates

Study Designs: Systematic reviews (with or without meta-analyses) of randomised controlled trials (RCTs).

Published Language: No restrictions were placed on language.

Exclusion Criteria:

- Abstracts, letters to the editor, commentary, case reports, editorials, personal communications, grey literature.
- Meta/umbrella/systematic overview of systematic reviews, narrative reviews, qualitative reviews, realist reviews, scoping reviews and review protocols.

Subgroups: Evidence was considered separately for populations with normal cognition and MCI. If sufficient evidence was available, evidence would be considered separately for key sub-groups of interest such as age (mid versus late life), gender, ethnicity, and/or socioeconomic profile.

2.3 Search strategy

Searches were conducted in English utilising the following bibliographic databases: Embase, MEDLINE, and Cochrane Library. The search strategies were built by the WHO librarian and tailored to each database to be consistent across all PICOs. Key terms related to “Cognitive Function,” “Diabetes Mellitus,” “Diabetes Interventions,” and “Systematic Review” were searched in the Title/Abstract Field, with relevant controlled vocabulary included in the search strings. Filters were used to limit results to articles in relevant publication formats (i.e., remove publications indexed as abstracts, editorials, etc) and in humans. No restrictions were made on published language. Searches were limited to articles published between 1st January 2020 and 23rd December 2024 (MEDLINE and Embase) / 31st December 2024 (Cochrane Library) (i.e., in the last five years, since the preceding systematic review was conducted). The full search strategy utilised for each database is reported in Appendix 1.

2.4 Selection of studies

The records retrieved were uploaded into Covidence for de-duplication, then assessed for eligibility by screening of titles and abstracts. Articles identified as potentially meeting the inclusion and exclusion criteria were then assessed for eligibility through full-text review. To ensure accuracy, two independent reviewers examined each article at both stages (i.e., title/abstract and full text), and any disagreements were resolved through discussion and

consensus between the two reviewers. A third reviewer was engaged if consensus could not be reached. A record of the search strategies, retrievals, and full-text exclusion reasons was maintained for transparency and replicability.

2.5 Data extraction and management

Data from eligible reviews was extracted into a document on Microsoft Word. An a-priori defined template was generated, then reflexively implemented (see section 2.8) in considering the needs of the review questions and evidence profile. Data extraction was performed by a single reviewer and checked by a second reviewer for accuracy. Any errors identified were flagged and remedied via consultation between the two extractors. The final Data Extraction Template is reported in Appendix 2. Included reviews were summarised narratively based on this extracted data.

2.6 Quality assessment

A Measurement Tool to Assess Systematic Reviews (AMSTAR)-II (13) was utilised to assess the quality of the identified reviews. Two independent reviewers conducted the quality appraisal blinded, and any discrepancies were resolved through discussion until consensus was achieved. In accordance with the guidance on the use of AMSTAR-II, articles were given an overall appraisal rating according to the following criteria:

- High (0–1 non-critical weaknesses)
- Moderate (>1 non-critical weakness)
- Low (1 critical flaw, with or without non-critical weaknesses)
- Critically low (>1 critical flaw, with or without non-critical weaknesses).

Included reviews were then ranked by AMSTAR-II rating, considering the number of critical flaws and/or non-critical weaknesses where overlap of ratings occurred. The quality assessment for the identified reviews is reported in Appendix 3.

2.7 Assessment of certainty of evidence

The Grading of Recommendations, Assessment, Development, and Evaluations (GRADE) System (14) was then applied. GRADE is used to robustly evaluate the level of certainty that can be held in available evidence, by considering the following domains:

- **Risk of Bias:** Level of internal validity. Evaluated based on the Risk of Bias assessment conducted in the primary review.
- **Inconsistency:** Level of variability (heterogeneity) in the results. Evaluated based on appropriate measurement of heterogeneity within the primary review (e.g., I^2 tests), or through consideration of the effect sizes and confidence intervals of the included studies.
- **Indirectness:** Level of fit to the present research question.
- **Imprecision:** Level of confidence in the estimates of effect. Evaluated by considering the sample size and confidence intervals.
- **Publication Bias:** Risk of/level of selective or one-sided reporting. When a systematic review contained a formal evaluation of publication bias (e.g., Egger's Test), this was used to evaluate this domain. When this was not available, the present researchers evaluated the comprehensiveness of the review in terms of search strategy, potential

bias created in study selection, and the time elapsed between search and publication, to discern whether publication bias had arisen.

Where multiple articles included in the present review addressed the same PICO, only one was selected for application of the GRADE system. The selection of articles was based on:

- Relative AMSTAR-II rating (with higher quality reviews prioritised over lower quality ones);
- The level of evidence according to the evidence hierarchy (with meta-analyses being prioritised over systematic reviews without meta-analyses);
- The “best fit” for a given PICO.

This process was repeated to ensure that available evidence on each of the outcomes of interest (incident dementia, incident MCI, cognitive function, quality of life, functional abilities, adverse events, drop-out rates) and relevant variations in intervention (e.g., glucose-lowering medication classes, various types of lifestyle interventions) were reported on.

The GRADE rating and decisions to downgrade certainty of evidence was conducted during regular meetings between three ERT members. The following thresholds were established in consultation with the WHO and other ERTs, to indicate benefit/harm for binary outcomes (i.e., dementia/MCI) or trivial/small/moderate/large effects for continuous outcomes:

- For binary outcomes: ≤ 10 fewer/more cases per 1 000 indicating trivial or no effect
- For continuous outcomes:
 - Trivial/no effect: standardised mean difference (SMD) $\leq |0.15|$
 - Small: SMD $|0.16| - |0.49|$
 - Moderate/large: SMD $\geq |0.50|$

Application of this system, resulted in the generation of a GRADE Evidence Profile for each intervention type (pharmacological and lifestyle/diet intervention) in each population group (cognitively normal, and with MCI) under investigation in the present review. These have been presented through Summary of Findings tables.

2.8 Deviations from the review protocol

The a priori developed Data Extraction Table was revised from the review protocol to be better suited for extracting review-level data items, and to support facilitation of PICO identification. The updated Data Extraction Table was implemented using a Microsoft Word document and has been included in Appendix 2. The implementation of the Data Extraction process was completed as per the review protocol. Additionally, there was insufficient evidence to consider additional subgroups of interest such as gender, ethnicity, age-group, or socioeconomic profile.

3. Results

3.1 Search results

The flow of records from retrieval to inclusion is depicted in Figure 1. Eighteen studies were included in the final systematic review, of which 9 contributed to the GRADE Profiles. Reports included in the systematic review but excluded from the GRADE Tables (n = 9) are noted in Section 3.1.2, with reasons for their exclusion supplied.

Fig WA.E.1 PRISMA 2020 Flow Diagram of review selection process

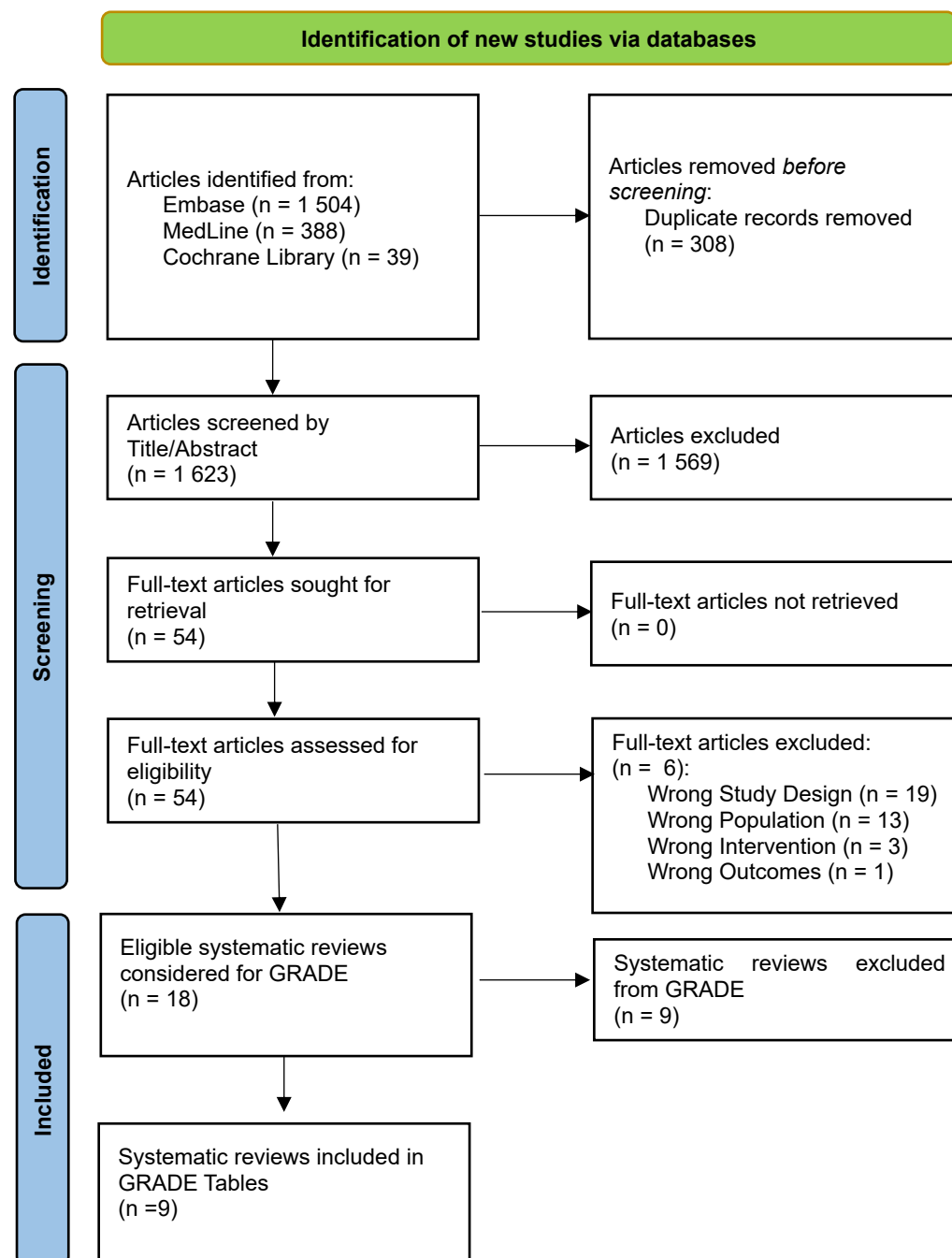


Table WA.E.1 Visual summary of PICO aspects addressed by the GRADED evidence

Population	Intervention	Critical outcomes			Other important outcomes	Comments
		Dementia	Incident MCI	Cognitive function		
Pharmacological interventions						
Adults with T2DM and normal cognition	Metformin ⁱ					No gradable intervention evidence identified
	Sulfonylurea ⁱⁱ					
	insulin					
	DPP-4i ⁱⁱⁱ	---	---	Meaningful benefit (small) ^d	--	
	GLP-1 RA	---	---	Meaningful benefit ^d		
	Intensive glucose control ^{iv}	No meaningful effect ^d	---	Little to no effect ^d		
	Intranasal insulin ^v	---	---	Meaningful benefit (small) ^d		
Adults with T2DM and MCI	Metformin ⁱ					No evidence identified
	Sulfonylurea ⁱⁱ					
	DPP-4i ⁱⁱⁱ	---	---	Meaningful benefit (large) ^d	---	
Lifestyle (Diet, Exercise)						
Adults with T2DM and normal cognition	Intensive diet and exercise	No meaningful effect ^d		---	---	
	Exercise	---	---	Meaningful benefit (large) ^d	Quality of Life	
Adults with T2DM and MCI	Exercise	---	---	Meaningful benefit (large) ^d	---	

Legend:

DPP-4i: dipeptidyl peptidase-4 inhibitor, GLP-1 RA: glucagon-like peptide-1 receptor agonists; MCI: mild cognitive impairment; T2DM: type 2 diabetes mellitus

ⁱ WHO recommended first-line treatment for T2DM (10, 15)

ⁱⁱ Second-generation sulfonylurea (preferably gliclazide) can be used as initial (first line) treatment when metformin is contraindicated or not tolerated. Sulfonylureas may cause weight gain or hypoglycaemia (10, 15).

ⁱⁱⁱ DPP4 inhibitors are the only third-line T2DM treatments for which gradable intervention evidence has been identified for their impact on cognition/dementia risk; no intervention evidence was identified for SGLT-2 inhibitors or thiazolidinedione (TZD). For observational evidence see section 3.4.5.

^{iv} “intensive glucose control” was achieving glycated haemoglobin (HbA1c) of <7% or a fasting glucose level of < 130 mg/dL.

^v not standard treatment for T2DM, evidence considered due to specific relevance for dementia prevention

Size of effect / threshold for meaningful benefit or harm:

For binary outcomes (i.e., dementia, MCI): >10 per 1 000 = meaningful benefit or meaningful harm; below that little to no meaningful effect.

For continuous outcomes (i.e., cognitive function): standardized mean differences (SMD) little to no effect up to 0.15; small 0.16 to 0.49; large 0.50 and above.

Certainty of evidence: red: very low certainty; orange: low certainty; yellow: moderate certainty; green: high certainty.

Quality of evidence (according to AMSTAR-II): a = high, b = moderate, c = low, d = very low

3.1.1 Included systematic reviews (n = 9)

No intervention evidence was identified under the review’s search parameters for WHO recommended first-line treatments of T2DM (i.e. metformin and sulfonylurea) and their effects on cognition, cognitive decline and dementia/MCI incidence

Pharmacological Agents in Cognitively normal T2DM population (in alphabetical order by first author)

García-Casares N, González-González G, de la Cruz-Cosme C, Garzón-Maldonado FJ, de Rojas-Leal C, Ariza MJ et al. Effects of GLP-1 receptor agonists on neurological complications of diabetes. Rev Endocr Metab Disord. 2023;24(4):655–72 (<https://doi.org/10.1007/s11154-023-09807-3>).

(AMSTAR-II rating: critically low)

Meng J, Yan R, Zhang C, Bai X, Yang X, Yang Y et al. Dipeptidyl peptidase-4 inhibitors alleviate cognitive dysfunction in type 2 diabetes mellitus. Lipids in Health and Disease. 2023;22(1):219 (<https://doi.org/10.1186/s12944-023-01985-y>). **(AMSTAR-II rating: critically low)**

Tang X, Cardoso MA, Yang J, Zhou JB, Simó R. Impact of Intensive Glucose Control on Brain Health: Meta-Analysis of Cumulative Data from 16 584 Patients with Type 2 Diabetes Mellitus. Diabetes Ther. 2021;12(3):765–79 (<https://doi.org/10.1007/s13300-021-01009-x>). **(AMSTAR-II rating: critically low)**

Wu S, Stogios N, Hahn M, Navagnanavel J, Emami Z, Chintoh A et al. Outcomes and clinical implications of intranasal insulin on cognition in humans: A systematic review and meta-analysis. PLoS One. 2023;18(6):e0286887 (<https://doi.org/10.1371/journal.pone.0286887>). **(AMSTAR-II rating: critically low) à NB while not a standard treatment for T2DM, intranasal insulin has been considered as intervention for GRADING due to**

specific relevance and potential benefit for dementia risk reduction at the request of GDG.

Pharmacological agents in T2DM population with MCI

Yuan Y, Zhang Y, Lei M, Guo X, Yang X, Ouyang C et al. Effects of DPP4 Inhibitors as Neuroprotective Drug on Cognitive Impairment in Patients with Type 2 Diabetes Mellitus: A Meta-Analysis and Systematic Review. *Int J Endocrinol*. 2024;2024:9294113 (<https://doi.org/10.1155/2024/9294113>). **(AMSTAR-II rating: critically low)**

Lifestyle Intervention in Cognitively Normal T2DM Population (in alphabetical order by first author)

Dyer AH, Briggs R, Mockler D, Gibney J, Kennelly SP. Non-pharmacological interventions for cognition in patients with Type 2 diabetes mellitus: a systematic review. *Qjm*. 2020;113(3):155–61 (<https://doi.org/10.1093/qjmed/hcz053>). **(AMSTAR-II rating: critically low)**

Sun Z, Liu H, Yan M, Zeng H, Hu Y, Tian X et al. The effect of multi-component exercise on cognition function in patients with diabetes: A systematic review and meta-analysis. *PLoS One*. 2024;19(6):e0304795 (<https://doi.org/10.1371/journal.pone.0304795>). **(AMSTAR-II rating: critically low)**

Zhang J, Tam WWS, Hounsri K, Kusuyama J, Wu VX. Effectiveness of Combined Aerobic and Resistance Exercise on Cognition, Metabolic Health, Physical Function, and Health-related Quality of Life in Middle-aged and Older Adults With Type 2 Diabetes Mellitus: A Systematic Review and Meta-analysis. *Arch Phys Med Rehabil*. 2024;105(8):1585–99 (<https://doi.org/10.1016/j.apmr.2023.10.005>). **(AMSTAR-II rating: critically low)**

Lifestyle Intervention in T2DM Population with MCI

Lu HH, Zhou Y, Chen C, Gu ZJ. Meta-analysis of the effect of exercise intervention on cognitive function in elderly patients with type 2 diabetes mellitus. *BMC Geriatr*. 2024;24(1):770. **(AMSTAR-II rating: critically low)**

Table WA.E.2 Summary of included systematic reviews

Study author	Study design	AMSTAR-II Rating	Outcome (synthesis method)	Intervention	Population (sample size: Intervention/ Control)	Comparator	Specific outcomes measure	Time Point of measurement
Pharmacological Interventions: Cognitively normal population								
Garcia-Casares et al. (16)	RCTs	Critically low	Cognitive function (narrative)	GLP-1 RAs	8 828 (4 456 / 4 372)	Placebo	MoCA; DSST	Median 5.4 years
Meng et al. (17)	RCTs	Critically low	Cognitive function (meta-analysis)	DPP-4i	4 768 (2 448 / 2 320)	Placebo / alternate glucose-lowering medicines	MMSE	6 months to 40 months
Tang et al. (18)	RCTs	Critically low	Cognitive function (meta-analysis)	Intensive glycaemic control	5 444 (2 751, 2 693)	Standard glycaemic control	MMSE; Comprehensive Assessment and Referral Evaluation (CARE)-Diagnostic scale	3.5 years to 10 years
			Incident dementia		11 140 (5 571 / 5 569)		First date of dementia medication use, hospital records, or significant global cognitive decline	Median 5 years
Wu et al. (19)	RCTs	Critically low	Cognitive function (meta-analysis)	Intranasal insulin (*)	N/A	Placebo	Standardised neuropsychological tests (Stroop Test, Delayed Word List Recall, Verbal Fluency Test)	8 weeks to 24 weeks
Pharmacological interventions: MCI population								
Yuan et al. (20)	RCTs	Critically low	Cognitive function (meta-analysis)	DPP-4i	5 388 (2 754 / 2 634)	Placebo or Alternate glucose-lowering medicines.	MMSE	24 weeks to 160 weeks.
Lifestyle interventions: cognitively normal population								
Dyer et al. (21)	RCTs	Critically low	Incident MCI (narrative)	Intensive lifestyle	3 802 (1 918 / 1 894)	Health education groups	Diagnosis by an expert panel, based on 3MSE result and	10-Year Follow-Up

			Incident dementia (narrative)				functional assessment	
Sun et al. (22)	RCTs	Critically low	Cognitive function (meta-analysis)	Exercise	1 193 (593 / 600)	Any (predominantly routine care or health administration)	Standardised neuropsychological tests (MMSE, 3MSE, MoCA, Stroop test)	12 weeks to 96 weeks
Zhang et al. (23)	RCTs	Critically low	Quality of Life (Meta-Analysis)	Exercise	867 (415 / 412)	Usual care	SF-36	12 weeks to 12 months
Lifestyle interventions: MCI population								
Lu et al. (24)	RCTs	Critically low	Cognitive function (meta-analysis)	Exercise (all types)	594 (299 / 295)	Routine care / Maintenance	MMSE; MoCA	3 months to 12 months

(*) not standard treatment for T2DM, evidence considered due to specific relevance for dementia prevention

3.1.2 Excluded systematic reviews (n = 9)

Table WA.E.3 Excluded systematic reviews

Study	PICO(s) covered	Reason for exclusion from GRADE
Jin et al. (25)	P: Adults with T2DM who are cognitively normal. I: DPP-4i / GLP1-RA C: Placebo/alternate anti-diabetic agents/glucose-lowering medicines O: Cognitive function, drop-out rates	Cognitive function Preference was given to Garcia-Casares et al. (16) (GLP1-RA) and Meng et al. (17) (DPP-4i) as it enabled consideration of the glucose-lowering classes separately, one of which was a higher-level evidence synthesis (meta-analysis). Drop-out rates Drop-out rates were reported narratively in the review; however not enough information was provided to appropriately GRADE. Details about drop-out rates have been included narratively in this evidence profile.
AMSTAR II: Low		
Cai et al. (26)	P: Adults with T2DM who are cognitively normal. I: Lifestyle (Exercise) C: Usual care / maintenance O: Cognitive function, adverse events.	Cognitive function Preference was given to Sun et al. (22) as it received a higher AMSTAR-II ranking and covered the same PICO. Adverse events Adverse events were reported narratively in the review; however not enough information was provided to appropriately GRADE. Details about adverse events have been included narratively in this evidence profile.
AMSTAR II: Critically low		
Cooke et al. (27)	P: Adults with T2DM. Cognition status not reported in review. I: Lifestyle (exercise)	Preference was given to Sun et al (22). as it received a higher AMSTAR-II ranking and covered the same PICO. Additionally, Cooke et al (27). was a poorer match for the PICO as it included

AMSTAR II: Critically low	C: Passive O: Cognitive function	participants aged less than 40.
Ding et al. (28)	P: Adults 40+ with T2DM and “Cognitive impairment” I: Lifestyle (aerobic exercise) C: Control or combined therapy O: Cognitive function	Preference was given to Lu et al (24). as it received a higher AMSTAR-II ranking and covered the same PICO. Lu et al. (24) was further deemed a better PICO match as Ding et al. (28) included articles that had used combined intervention (e.g., lifestyle and pharmacological) as their comparator, and it was unclear how the review defined “cognitive impairment” as a population inclusion criterion.
AMSTAR II: Critically low		
Su et al. (29)	P: Adults 40+ with T2DM, and “Diabetes Associated Cognitive Decline” I: Traditional Chinese medicine C: Alternate hypoglycaemic medicines O: Cognitive function, adverse events	Inadequate PICO Match. This review has been included in the “additional information” section however due to lack of representation of traditional Chinese medicine in other included reviews.
AMSTAR II: Low		
Nijssen et al. (30)	P: Adults 40+ with T2DM. Cognition status not reported in review. I: Intranasal insulin C: Placebo O: Cognitive function	Preference was given to Wu et al. (19) as it covered the same PICO but had a higher AMSTAR-II ranking.
AMSTAR II: Critically low		
Crabtree et al. (31)	P: Adults 40+ with T2DM. Cognition status not reported in review. I: Intensive glycaemic control C: Standard care O: Cognitive function	Preference was given to Tang et al. (18) as it covered the same PICO, was of a similar AMSTAR-II Ranking, and held a higher-level evidence synthesis (meta-analysis compared to narrative).
AMSTAR II: Critically low		
Chai et al. (32)	P: Adults with T2DM. With mixed cognition at baseline.	Preference was given to Meng et al. (17) as it covered the same PICO but held a higher AMSTAR-II Ranking.

	I: Incretin-Based (GLP-1 RAs and DPP-4i). C: Placebo/alternate anti-diabetic agents/glucose-lowering medicines O: Cognitive function	
AMSTAR II: Critically low		
Slouha et al. (33)	P: Adults with T2DM. Cognition status not reported in review. I: Glucose-lowering medicines C: Various/uncontrolled O: Cognitive function, incident dementia	This report was excluded from the GRADE Evidence Profiles as it was predominated by observational and cohort studies. The few RCTs that were narratively summarised in this review were better represented by other reports with higher AMSTAR-Ratings.
AMSTAR II: Critically low		

3.2 PICO table

Table WA.E.4 PICO table

Serial number	Population	Intervention vs comparison	Outcomes	Intervention sub-type: Systematic reviews used for GRADE	Justification for systematic review used
1	Adults 40+ with T2DM and normal cognition	Pharmacological anti-diabetes agents/ glucose-lowering medicines vs. usual care / placebo / no intervention	Dementia	Intensive Glycaemic Control: Tang X, Cardoso MA, Yang J, Zhou JB, Simó R. Impact of Intensive Glucose Control on Brain Health: Meta-Analysis of Cumulative Data from 16 584 Patients with Type 2 Diabetes Mellitus. <i>Diabetes Ther.</i> 2021;12(3):765–79 (https://doi.org/10.1007/s13300-021-01009-x).	Most recent and highest quality systematic review that includes evidence on intensive versus standard glycaemic control and incident dementia in adults with presumed normal cognition. AMSTAR-II rating is critically low.
			Incident MCI	No relevant systematic review available	N/A
			Cognitive function	GLP1-RAs: García-Casares N, González-González G, de la Cruz-Cosme C, Garzón-Maldonado FJ, de Rojas-Leal C, Ariza MJ et al. Effects of GLP-1 receptor agonists on neurological complications of diabetes. <i>Rev Endocr Metab Disord.</i> 2023;24(4):655–72 (https://doi.org/10.1007/s11154-023-09807-3).	Most recent and highest quality systematic review that includes evidence on GLP1-RAs versus a comparator on cognitive function in adults with presumed normal cognition. AMSTAR-II rating is

critically low.

Narrative reporting of 1 relevant RCT is included.

DPP-4i: Meng J, Yan R, Zhang C, Bai X, Yang X, Yang Y et al. Dipeptidyl peptidase-4 inhibitors alleviate cognitive dysfunction in type 2 diabetes mellitus. *Lipids in Health and Disease*. 2023;22(1):219 (<https://doi.org/10.1186/s12944-023-01985-y>).

Most recent and highest quality systematic review that includes evidence on DPP-4is versus a comparator on cognitive function in adults with normal cognition. **AMSTAR-II rating is critically low.**

Meta-analysis of 3 relevant RCTs is included.

Intensive Glycaemic Control: Tang X, Cardoso MA, Yang J, Zhou JB, Simó R. Impact of Intensive Glucose Control on Brain Health: Meta-Analysis of Cumulative Data from 16 584 Patients with Type 2 Diabetes Mellitus. *Diabetes Ther*. 2021;12(3):765–79 (<https://doi.org/10.1007/s13300-021-01009-x>).

Most recent and highest quality systematic review that includes evidence on intensive versus standard glycaemic control and cognitive function in adults with presumed normal cognition. **AMSTAR-II rating is critically low.**

					Meta-analysis of 4 relevant RCTs is included
				Intranasal Insulin: Wu S, Stogios N, Hahn M, Navagnanavel J, Emami Z, Chintoh A et al. Outcomes and clinical implications of intranasal insulin on cognition in humans: A systematic review and meta-analysis. PLoS One. 2023;18(6):e0286887 (https://doi.org/10.1371/journal.pone.0286887).	Most recent and highest quality systematic review that includes evidence on intranasal insulin versus placebo on cognitive function in adults with presumed normal cognition. While not a treatment for T2DM, evidence considered due to specific relevance for dementia prevention. AMSTAR-II rating is critically low.
			Quality of Life	No relevant systematic review available	N/A
			Functional levels (ADL)	No relevant systematic review available	N/A
			Adverse events	No relevant systematic review available	N/A
2	Adults 40+ with T2DM and normal	lifestyle interventions vs.	Dementia	Intensive Lifestyle: Dyer AH, Briggs R, Mockler D, Gibney J, Kennelly SP. Non-pharmacological	Most recent and highest quality

cognition	usual care / placebo / no intervention	interventions for cognition in patients with Type 2 diabetes mellitus: a systematic review. Qjm. 2020;113(3):155–61 (https://doi.org/10.1093/qjmed/hcz053)	systematic review that includes evidence on intensive lifestyle (diet and exercise) intervention versus control on incident dementia in adults with presumed normal cognition. AMSTAR-II rating is critically low. Narrative reporting of 1 relevant RCT is included.
Incident MCI		Intensive lifestyle: Dyer AH, Briggs R, Mockler D, Gibney J, Kennelly SP. Non-pharmacological interventions for cognition in patients with Type 2 diabetes mellitus: a systematic review. Qjm. 2020;113(3):155–61 (https://doi.org/10.1093/qjmed/hcz053)	Most recent and highest quality systematic review that includes evidence on intensive lifestyle (diet and exercise) intervention versus control on incident MCI in adults with presumed normal cognition. AMSTAR-II rating is critically low. Narrative reporting of 1 relevant RCT is included.
Cognitive		Exercise: Sun Z, Liu H, Yan M, Zeng H, Hu Y, Tian X et al.	Most recent and

function	The effect of multi-component exercise on cognition function in patients with diabetes: A systematic review and meta-analysis. PLoS One. 2024;19(6):e0304795 (https://doi.org/10.1371/journal.pone.0304795)	highest quality systematic review that includes evidence on exercise intervention versus routine care on cognitive function in adults with presumed normal cognition. AMSTAR-II rating is critically low. Meta-analysis of 12 relevant RCTs is included.
Quality of Life	Exercise: Zhang J, Tam WWS, Hounsri K, Kusuyama J, Wu VX. Effectiveness of Combined Aerobic and Resistance Exercise on Cognition, Metabolic Health, Physical Function, and Health-related Quality of Life in Middle-aged and Older Adults With Type 2 Diabetes Mellitus: A Systematic Review and Meta-analysis. Arch Phys Med Rehabil. 2024;105(8):1585–99 (https://doi.org/10.1016/j.apmr.2023.10.005).	Most recent and highest quality systematic review that includes evidence on exercise intervention versus routine care on quality of life in adults with presumed normal cognition. AMSTAR-II rating is critically low. Meta-analysis of 5 relevant RCTs is included.
Functional levels (ADL)	No relevant systematic review available	N/A

3	Adults 40+ with T2DM and MCI	Pharmacological anti-diabetes agents/ glucose-lowering medicines vs. usual care / placebo / no intervention	Adverse events	No relevant systematic review available	N/A
			Drop-out rates	No relevant systematic review available	N/A
			Dementia	No relevant systematic review available	N/A
			Incident MCI	No relevant systematic review available	N/A
			Cognitive function	DPP-4i: Yuan Y, Zhang Y, Lei M, Guo X, Yang X, Ouyang C et al. Effects of DPP4 Inhibitors as Neuroprotective Drug on Cognitive Impairment in Patients with Type 2 Diabetes Mellitus: A Meta-Analysis and Systematic Review. Int J Endocrinol. 2024;2024:9294113 (https://doi.org/10.1155/2024/9294113).	Most recent and highest quality systematic review that includes evidence on DPP-4i treatment compared to a control in adults with cognitive impairment. AMSTAR-II rating is critically low.
					Meta-analysis of 9 relevant RCTs is included.
			Quality of Life	No relevant systematic review available	N/A
			Functional levels (ADL)	No relevant systematic review available	N/A
			Adverse events	No relevant systematic review available	N/A
			Drop-out rates	No relevant systematic review available	N/A
4	Adults 40+ with T2DM and MCI	lifestyle interventions vs. usual care / placebo / no intervention	Dementia	No relevant systematic review available	N/A
			Incident MCI	No relevant systematic review available	N/A
			Cognitive function	Exercise: Lu HH, Zhou Y, Chen C, Gu ZJ. Meta-analysis of the effect of exercise intervention on cognitive function in elderly patients with type 2 diabetes mellitus. BMC Geriatr. 2024;24(1):770.	Most recent and highest quality systematic review that includes evidence on the impact of exercise on

cognitive function in adults with MCI.
AMSTAR-II rating is critically low.

Meta-analysis of 6 relevant RCTs is included.

Quality of Life	No relevant systematic review available	N/A
Functional levels (ADL)	No relevant systematic review available	N/A
Adverse events	No relevant systematic review available	N/A
Drop-out rates	No relevant systematic review available	N/A

3.3 GRADE evidence profiles with narrative summaries

3.3.1 Glucose-lowering medicines/anti-diabetes agents in cognitively normal T2DM patients

Population: Adults aged 40+ with T2DM and who are cognitively normal.

Intervention: Glucose-lowering medicines

Comparator: Usual care / placebo / no intervention

Table WA.E.4a GRADE Summary of findings- Glucose-lowering medicines/anti-diabetes agents vs usual care/placebo/no intervention in adults with T2DM and normal cognition

Outcome	Intervention	Review results & measurement	Absolute effect estimates		Certainty of evidence	Summary
			Intervention	Comparator		
Incident Dementia	Intensive Glucose Control	RR = 1.27, 95% CI [0.87,1.85]	61/5 571	48/5 569	Moderate Due to serious imprecision ^{a,b}	Intensive glucose control likely results in little to no difference in incident dementia compared to standard glucose control.
		Based on data from 11 140 participants in 1 RCT ¹ Follow up: median 5 years Assessed with first date dementia medication use, hospital records, significant global cognitive decline.	2 more per 1 000 (from 1 fewer to 7 more)			
Cognitive Function	DPP-4i	Based on data from 4 768 participants in 3 RCTs ²	2 448	2 320	Very low Due to serious risk of bias ^g , very serious inconsistency ^h , serious indirectness ⁱ , and serious imprecision ^j	The evidence is very uncertain about the effect of DPP-4i treatment on cognitive function.
		Follow-up range: 6 months to 40 months Assessed with: MMSE	MD = 1.29, 95% CI [-0.03, 2.60]			
	GLP-1 RA	HR = 0.93, 95% CI [0.85, 1.02]	905/4 456	944/4 372	Very low Due to serious risk of bias ^k , serious indirectness ^l , and serious imprecision ^m	The evidence is very uncertain about the effect of GLP-1RA on cognitive decline. ⁿ
		Based on data from 8 828 participants in 1 RCT ³ Follow-up: median 5.4 years Assessed with: MoCA and DSST	13 fewer per 1 000 (from 29 fewer to 4 more)			

Intensive Glucose Control	Based on data from 5 444 participants in 4 RCTs ¹ Follow-up: range 3.5 years to 10 years Assessed with: MMSE, Comprehensive Assessment and Referral Evaluation-Diagnostic Scale, additional various neuropsychological batteries.	2 751 b = -0.03, 95% CI [-0.05, -0.02]	2 693	Low Due to very serious indirectness ^c	Intensive glucose control may result in little to no difference in cognitive function compared to standard glucose control. ^d
Intranasal Insulin (*) *While not a treatment for T2DM, evidence considered due to specific relevance for dementia prevention	Based on data from 165 participants in 3 RCTs ⁴ Follow-up range: 8 weeks to approx. 40 weeks Assessed with: various standardized neuropsychological tests.	81 SMD = 0.18, 95% CI [-0.11, 0.48]	84	Very low Due to very serious indirectness ^e and very serious imprecision ^f	The evidence is very uncertain about the effect of Intranasal Insulin on cognitive function.

^a **Imprecision:** serious. OIS not met, CI includes no effect.

^b **Interpretation Note:** As information is based on only 1 RCT, inconsistency could not appropriately be assessed. Evidence has not been downgraded for this but should be interpreted with caution.

^c **Indirectness:** very serious. Due to concerns regarding indirectness of the intervention and comparators. The interventions in the primary articles include multi-factorial (e.g., of multiple cardio-metabolic risk factors, lifestyle) and telemedicine, which introduces potential for confounding. There is additionally overlap of "standard" and "intensive" blood pressure targets between the included studies. For example, in Koekkoek the routine care group initially followed guideline targets of <8.5% HbA1c, before transitioning to <7.0% HbA1c when guidelines changed. This 7.0% is equivalent to the targets of the intervention conditions in other included studies. Additionally, one of the included studies is an observational ancillary of a prior RCT which introduces concern. Downgraded by two levels due to the likelihood of these alternative factors influencing (be it accounting for, or obscuring) effects.

^d **Interpretation Note:** The systematic review conducted a meta regression analysis with no indication of standardisation. Due to the range of measurement tools utilised to measure cognitive function across studies, it is not suitable to convert the observed b into a SMD, and therefore the thresholds for meaningful effect adopted in the present review cannot be applied. A contextual judgement has therefore been made of the observed effect as "little to no effect." Additionally, as the observed CI does not cross the line of no effect, is narrow, and is based on a large sample size, evidence was not downgraded for imprecision.

^e **Indirectness:** very serious. Population is not a direct match to the PICO, one of the RCTs in the meta-analysis has a population without T2DM and with a mean age <40. This population differs meaningfully in metabolic profile and treatment context.

^f **Imprecision:** very serious. The 95% CI is very wide, including potential for trivial harm and appreciable benefit. Sample size substantially < OIS.

^g **Risk of bias:** serious. RCTs at unclear to high risk of bias.

^h **Inconsistency:** very serious. There may be high levels of heterogeneity ($I^2 = 98\%$), supported by inconsistent overlap of CIs between studies.

ⁱ **Indirectness:** serious. Population is not a direct match to the PICO, one of the RCTs has a population with MCI, contributes a relatively small portion of participants to the meta-analysis.

^j **Imprecision:** serious. CI includes both meaningful benefit and little to no effect. Conversion of the reported MD to a SMD was conducted based on a conservative typical MMSE SD (≈ 3.0) (34, 35) for cognitively normal adults in middle adulthood or older, resulting in an approximate SMD = 0.43, 95%CI [-0.01, 0.87].

^k **Risk of bias:** serious. Included study at unclear risk of bias.

^l **Indirectness:** The sample included participants with MoCA scores at baseline indicative of potential MCI.

^m **Imprecision:** serious. Sample size > OIS. 95% CI includes both little to no harm and appreciable benefit

ⁿ **Interpretation Note:** As information is based on only 1 RCT, inconsistency could not appropriately be assessed. Evidence has not been downgraded for this but should be interpreted with caution.

References

1. Tang X, Cardoso MA, Yang J, Zhou JB, Simó R. Impact of Intensive Glucose Control on Brain Health: Meta-Analysis of Cumulative Data from 16 584 Patients with Type 2 Diabetes Mellitus. *Diabetes Ther.* 2021;12(3):765–79 (<https://doi.org/10.1007/s13300-021-01009-x>).
2. Meng J, Yan R, Zhang C, Bai X, Yang X, Yang Y et al. Dipeptidyl peptidase-4 inhibitors alleviate cognitive dysfunction in type 2 diabetes mellitus. *Lipids in Health and Disease.* 2023;22(1):219 (<https://doi.org/10.1186/s12944-023-01985-y>).
3. García-Casares N, González-González G, de la Cruz-Cosme C, Garzón-Maldonado FJ, de Rojas-Leal C, Ariza MJ et al. Effects of GLP-1 receptor agonists on neurological complications of diabetes. *Rev Endocr Metab Disord.* 2023;24(4):655–72 (<https://doi.org/10.1007/s11154-023-09807-3>).
4. Wu S, Stogios N, Hahn M, Navagnanavel J, Emami Z, Chintoh A et al. Outcomes and clinical implications of intranasal insulin on cognition in humans: A systematic review and meta-analysis. *PLoS One.* 2023;18(6):e0286887 (<https://doi.org/10.1371/journal.pone.0286887>).

Narrative summaries

1. *Tang et al. (18): Intensive versus standard glycaemic control*

A meta-analysis completed by Tang et al. (18) appraised the impact of intensive compared to standard glucose control on outcomes related to brain health in T2DM patients. Intensive Control was defined in the review as pharmacological glycaemic control that achieved glycated haemoglobin (HbA1c) less than 7%, or fasted glucose level <130mg/dL. Four databases (PubMed, Embase, Web of Science, and Clinicaltrial.gov) were searched from inception to July 2020, to identify relevant RCTs. Six RCTs (n = 16 584) were ultimately included in the review, with outcomes reported for incident dementia and cognitive function. The cognition status of participants was not directly identified as part of the review's eligibility criteria, but MMSE scores of the included studies at baseline are indicative of cognitively normal participants predominating the sample.

One RCT (36) (n = 11 140) compared risk of incident dementia between the intensive (n = 5 571) and standard (n = 5 569) glucose control groups. The review reported the following risk ratio: RR = 1.27, 95% CI [0.87, 1.85]. When this was converted to an absolute effect based on event rates sourced from the primary article, the effect was deemed trivial (2 per 1 000, from 1 fewer to 7 more) at a median follow-up of 5 years. *

Four of the RCTs (37-40) (n = 5 444) were included in the meta-analysis to evaluate the effects of intensive compared to standard glucose control on cognitive function. Pooled estimates indicated a small decline in cognitive function for the intensive control group (n = 2 751) compared to the standard control group (n = 2 693) (b = -0.03, 95% CI [-0.05, -0.02]). Heterogeneity was low in this analysis ($I^2 = 19.9\%$), and all included trials were of a high quality when considered against the Newcastle-Ottawa Quality Assessment Scale.

AMSTAR-II quality rating of this review: critically low

GRADE interpretation for the **outcome of dementia**

The evidence suggests with moderate certainty that intensive glucose control likely results in little to no difference in incident dementia compared to standard glucose control in adults with T2DM and healthy cognition, at a median follow-up of 5 years,

GRADE interpretation for the **outcome of cognitive function**

The direction of effect suggests with low certainty that intensive glucose control results in little to no effect on cognitive function compared to standard glucose control in adults with T2DM and healthy cognition, at a follow up period of 3.5 years to 10 years.

The review team wishes to note that whilst the methodology for the present evidence profiles is based on an umbrella review format, concerns regarding the quality of this review (in addition to the critically low AMSTAR-II rating) were identified and reinforced by the WHO Technical Team. As such, in completing the GRADE tables and interpretations, the primary articles were sourced to support accurate representation of the primary studies (for example: regarding the nature of the interventions and comparators), and interpretation notes have been provided where appropriate. It should be noted however that these concerns regarding the quality/reporting in this review may extend beyond what is immediately identifiable, and as such the above results should be interpreted with caution.

2. Meng et al. (17): DPP-4i

In the systematic review and meta-analysis conducted by Meng et al. (17) the literature was searched to identify RCTs and observational studies that compared administration of dipeptidyl peptidase-4 inhibitors to standard care in T2DM patients. Searches were run from inception to September 2023 in 4 databases (PubMed, Embase, Cochrane Library, Web of Science). Six studies met the inclusion criteria for the review. Three of these included studies (41-43) were RCTs that evaluated the effectiveness of DPP-4i treatment on cognitive function, as measured through the Mini-Mental State Examination (MMSE). Risk of bias in these studies ranged from unclear to high.

When pooled analyses were conducted to examine cognitive function after DPP-4i treatment (n = 2 448) compared to a control (n = 2 320), results favoured the intervention (MD = 1.29, 95% CI [-0.03, 2.60]). However, the confidence interval included the possibility of little to no effect.

AMSTAR-II quality rating of this review: critically low

GRADE interpretation for the **outcome of cognitive function**

Due to the very low certainty evidence, we are unable to determine whether DPP-4i treatment impacts cognitive function compared to usual care / placebo / no intervention in adults with healthy cognition and T2DM at a follow up period of 6 to 40 months.

3. García-Casares. (16): GLP1-RA versus control

A systematic review was conducted by Garcia-Casares et al. (16) to identify the available evidence for the use of glucagon-like peptide receptor agonists (GLP-1-RA) on management of neurological complications associated with T2DM (e.g., stroke, peripheral neuropathy, cognitive impairment). The investigators searched three databases (MedLine, Cochrane, and Scopus) for articles published from

inception to January 2022. Nineteen clinical trials were included in the review, of which two were relevant RCTs for our PICO (i.e., RCTs with cognitive function as an outcome for T2DM patients with healthy cognition). Quantitative details were not comprehensively provided in the review, so the primary articles were consulted. Of the two RCTs, only one then presented data with a statistical comparison between the intervention and placebo groups and thus this RCT alone (44) informed the GRADE table. This RCT is summarised narratively below.

One RCT conducted by Cukierman-Yaffe et al., (44) compared the administration of dulaglutide (n = 4 456) to placebo (n = 4 372) in risk of cognitive decline, measured by the Montreal Cognitive Assessment (MoCA) and Digit Symbol Substitution Test (DSST). Cognitive decline was presented as a binary outcome; whereby decline was defined as first instance of MoCA or DSST score more than 1.5 SDs below the mean baseline score of the sample. Cukierman-Yaffe et al., reported that this threshold was prespecified, and was set to “exceed any random interindividual and intraindividual variation in both cognitive test scores and the change in scores with time.”(44)

The hazard ratio identified favoured the intervention: HR = 0.93, 95% CI [0.85, 1.02]. When this was converted to an absolute effect based on event rates sourced from the primary article, the effect indicated meaningful benefit of the intervention (13 fewer instances of cognitive decline per 1 000, from 29 fewer to 4 more) at a median follow-up of 5.4 years.

AMSTAR-II quality rating of this review: critically low

GRADE interpretation for the **outcome of cognitive function**

Due to the very low certainty evidence, we are unable to determine whether GLP-1 RA treatment impacts cognitive function compared to usual care / placebo / no intervention in adults with T2DM and healthy cognition at a median follow-up of 5.4 years.

4. Wu et al. (19): Intranasal Insulin

In the systematic review conducted by Wu et al. (19) the effects of intranasal insulin in a range of populations, including cognitively healthy individuals with T2DM, was explored. While not a treatment for T2DM, evidence for intranasal insulin was considered due to specific relevance for dementia prevention. The primary outcome of interest was cognitive function. Seven databases were searched from inception to February 2023. This review had a critically low AMSTAR-II rating, and significant reporting issues including a lack of clarity around the actual number of studies it included. As such, information we cannot be certain about has been omitted from this summary and data used to inform this evidence profile has been cross-checked with the primary studies. Three of the RCTs (45-47) included in the review examined the impact of intranasal insulin on cognitive function in participants with metabolic syndrome compared to placebo. These studies were at a low or unclear risk of bias.

When pooled analyses were conducted to examine intranasal insulin usage (n = 81) compared to placebo (n = 84), a small effect favouring the intervention was observed (SMD = 0.18, 95% CI [-0.11, 0.48]). Notably, one of the studies included in this meta-analysis (45) consisted of participants without T2DM and who had a mean age <40. These participants contributed a minority to the meta-analytic sample, however due to their markedly different metabolic profile and treatment context, it may still

impact certainty of results particularly in the context of an overall small sample size. Evidence has been downgraded accordingly. There was no statistical evidence of heterogeneity ($I^2 = 0\%$).

AMSTAR-II quality rating of this review: critically low

GRADE interpretation for the **outcome of cognitive function**

Due to the very low certainty evidence, we are unable to determine whether intranasal insulin impacts cognitive function compared to usual care / placebo / no intervention in adults with T2DM and healthy cognition at a follow up period of 8 to approximately 40 weeks.

3.3.2 Glucose-lowering medicines in T2DM patients with MCI

Population: Adults aged 40+ with T2DM and MCI

Intervention: Glucose-lowering medicines

Comparator: Usual care / placebo / no intervention

Table WA.E.4b GRADE summary of findings- glucose-lowering medicines vs usual care/placebo/no intervention in adults with T2DM and MCI

Outcome	Intervention	Review results & measurement	Absolute effect estimates		Certainty of evidence	Summary
			Intervention	Comparator		
Cognitive Function	DPP-4i	Based on data from 5 388 participants in 9 RCTs ¹ Follow up: range 24 weeks to 160 weeks Assessed with: MMSE	2 754 SMD = 0.99, 95% CI [0.59, 1.38]	2 634	Very Low Due to serious risk of bias ^a , serious inconsistency ^b , and serious indirectness ^c	The evidence is very uncertain about the effect of DPP-4i treatment on cognitive function.

^a **Risk of Bias:** serious. Included studies range from low to high risk of bias.

^b **Inconsistency:** serious. Heterogeneity may be considerable ($I^2 = 97\%$). Large variation of point estimates and minimal overlap of confidence intervals. The review team discussed the potential to downgrade by two levels due to the severity of potential inconsistency; however, this did not eventuate as the observed differences are between moderate and large treatment effects in the same direction.

^c **Indirectness:** serious. Comparator indirectness: included studies have both placebo ($n = 2$) and alternate glucose-lowering medication/anti-diabetic agent ($n=7$) as comparators. Population indirectness: some participants without MCI are included in the meta-analytic sample, however these represent the minority.

References

1. Yuan Y, Zhang Y, Lei M, Guo X, Yang X, Ouyang C et al. Effects of DPP4 Inhibitors as Neuroprotective Drug on Cognitive Impairment in Patients with Type 2 Diabetes Mellitus: A Meta-Analysis and Systematic Review. Int J Endocrinol. 2024;2024:9294113 (<https://doi.org/10.1155/2024/9294113>).

Narrative summaries

Yuan et al. (20): DPP-4i

There was minimal evidence identified for the use of glucose-lowering medicines to reduce the risk of dementia and cognitive decline in individuals with T2DM and MCI. One systematic review contributed towards this GRADE Table.

Yuan et al. (20) conducted a systematic review and meta-analysis to evaluate the impact of DPP-4i for treatment of cognitive function. The review aimed to identify publications involving this treatment within individuals who had cognitive impairment caused by T2DM. Three databases were searched (PubMed, CNKI, Cochrane Library), resulting in the inclusion of 10 RCTs with low to high risk of bias. Whilst not directly reported in the review, analysis of the reference list indicated the included RCTs largely comprised of populations with MCI. Meta-Analysis of nine included RCTs (41-43, 48-53) with the MMSE as a measure of cognitive function, revealed a higher MMSE score in the intervention group (n = 2 754) treated with DPP-4i compared to control groups (n = 2 634) treated with alternate glucose-lowering medicines or placebo (SMD = 0.99; 95% CI [0.59, 2.38]). High heterogeneity was observed in the results; however, this was robust to sub-group analyses of treatment duration and recipient age.

AMSTAR-II quality rating of this review: critically low

GRADE interpretation for the outcome of cognitive function

Due to the very low certainty evidence, we are unable to determine whether DPP-4i treatment impacts cognitive function compared to usual care / placebo / no intervention in adults with T2DM and MCI at a follow-up range of 24 to 160 weeks.

3.3.3 Lifestyle intervention in cognitively normal T2DM patients

Population: Adults aged 40+ with T2DM and who are cognitively normal.

Intervention: Lifestyle (diet, exercise)

Comparator: Usual care / placebo / no intervention

Table WA.E.4c GRADE summary of findings- lifestyle/diet interventions vs usual care/placebo/no intervention in adults with T2DM and normal cognition

Outcome	Intervention	Review results & measurement	Absolute effect estimates		Certainty of evidence	Summary
			Intervention	Comparator		
Incident Dementia	Intensive Diet and Exercise	RR = 0.98, 95% CI [0.61, 1.57] ^a Based on data from 3 802 participants in 1 RCT ¹ Follow up: mean 11.4 years Assessed with: expert adjudication, based on 3MSE and Functional Assessment Questionnaire.	34 / 1 918	34 / 1 884	Very Low Due to serious risk of bias ^b and very serious imprecision ^c	The evidence is very uncertain about the effect of intensive diet and exercise intervention on incident Dementia. ^d
			Difference: 0 fewer per 1 000 (from 7 fewer to 10 more)			
Incident MCI	Intensive Diet and Exercise	RR = 1.00, 95% CI [0.78, 1.27] ^a Based on data from 3 802 participants in 1 RCT ¹ Follow up: mean 11.4 years Assessed with: expert adjudication, based on 3MSE and Functional Assessment Questionnaire.	122 / 1 918	124 / 1 884	Very Low Due to serious risk of bias ^b and very serious imprecision ^e	The evidence is very uncertain about the effect of Intensive diet and exercise on Incident MCI ^d .
			Difference: 0 fewer per 1 000 (from 14 fewer to 18 more)			
Cognitive Function	Exercise	Based on data from 1 193 participants in 12 RCTs ²	593	600	Very Low Due to serious inconsistency ^f , serious indirectness ^g ,	The evidence is very uncertain about the effect of exercise on
			SMD = 0.91 95% CI [0.28, 1.54] ^c			

		Follow up: range 12 weeks to 96 weeks			and serious imprecision ^h	cognitive function
		Assessed with: MMSE, 3MSE, MoCA, Stroop Test				
Quality of Life – Physical	Exercise	Based on data from 867 participants in 5 RCTs ³	455	412	Very Low	The evidence is very uncertain about the effect of exercise on mental components of quality of life.
		Follow up: range 12 weeks to 12 months	MD = 4.17 95% CI [0.86, 7.48]		Due to serious risk of bias ⁱ , serious inconsistency ^j , and strong suspicion of publication bias ^k .	
		Assessed with: SF-36 Physical Component Score				
Quality of Life - Mental	Exercise	Based on data from 867 participants in 5 RCTs ³	455	412	Very Low	The evidence is very uncertain about the effect of exercise on mental components of quality of life.
		Follow up: range 12 weeks to 12 months	MD = 1.74 95% CI [-3.46, 6.94]		Due to serious risk of bias ⁱ , serious inconsistency ^l , serious imprecision ^m and strong suspicion of publication bias ^k	
		Assessed with: SF-36 Physical Component Score				

^a **Interpretation Note:** The noted RR has been calculated by the review team. The Dyer review only reported incident dementia and incident MCI prevalence rates in % form. To ensure accurate interpretation, the review team referred to the primary article to identify event rates and confirm sample sizes of each group for inclusion in this evidence profile.

^b **Risk of Bias:** serious. Multiple domains at high risk of bias.

^c **Imprecision:** very serious. OIS not met, CI includes no effect, and CI includes both little to no effect and meaningful harm.

^d **Interpretation Note:** As information is based on only 1 RCT, inconsistency could not appropriately be assessed. Evidence has not been downgraded for this, but results should be interpreted with caution.

^e **Imprecision:** very serious. OIS not met, CI includes no effect, and CI includes both meaningful benefit and meaningful harm.

^f **Inconsistency:** serious. There may be high levels of heterogeneity ($I^2 = 95\%$).

^g **Indirectness:** serious. Population indirectness: review did not exclude participants with cognitive impairment or dementia at baseline. Review of characteristics of included studies indicates various cognitive states may be represented.

^h **Imprecision:** serious. CI includes small to large favourable effect, in a relatively small sample size. Uncertainty in the magnitude of effect may be important.

ⁱ **Risk of Bias:** serious. Most included articles were rated as high risk of bias in 2+ domains.

^j **Inconsistency:** serious. There may be high levels of heterogeneity ($I^2 = 84\%$). Downgraded by only one level as most confidence intervals overlap.

^k **Publication Bias:** Publication bias is suspected due to results only being available for a small number of small trials, which predominantly favour intervention. There were too few studies to assess publication bias statistically.

^l **Inconsistency:** serious. There may be high levels of heterogeneity ($I^2 = 90\%$). Downgraded by only one level as most confidence intervals overlap.

^m **Imprecision:** serious. The CI is wide and includes both harm and benefit.

References

1. Dyer AH, Briggs R, Mockler D, Gibney J, Kennelly SP. Non-pharmacological interventions for cognition in patients with Type 2 diabetes mellitus: a systematic review. *Qjm*. 2020;113(3):155–61 (<https://doi.org/10.1093/qjmed/hcz053>).
2. Sun Z, Liu H, Yan M, Zeng H, Hu Y, Tian X et al. The effect of multi-component exercise on cognition function in patients with diabetes: A systematic review and meta-analysis. *PLoS One*. 2024;19(6):e0304795 (<https://doi.org/10.1371/journal.pone.0304795>).
3. Zhang J, Tam WWS, Hounsri K, Kusuyama J, Wu VX. Effectiveness of Combined Aerobic and Resistance Exercise on Cognition, Metabolic Health, Physical Function, and Health-related Quality of Life in Middle-aged and Older Adults With Type 2 Diabetes Mellitus: A Systematic Review and Meta-analysis. *Arch Phys Med Rehabil*. 2024;105(8):1585–99 (<https://doi.org/10.1016/j.apmr.2023.10.005>).

Narrative summaries

1. Dyer et al. (21): Intensive diet and exercise

Dyer et al. (21) conducted a systematic review to evaluate the effectiveness of non-pharmacological interventions on cognition in individuals with T2DM. Searches were conducted across four databases, and articles included if they evaluated an exercise, dietary, or cognitive training intervention in comparison to standard treatment on cognitive based outcomes (including incident MCI and incident dementia). Two studies were included in the review and summarised narratively. One of these RCTs, the LookAHEAD Trial (54) ($n = 3\,802$) has been included in the GRADE Tables. This trial was at a high risk of bias, and examined the effects of a 10-year intensive lifestyle intervention (involving a diet and exercise regime) in individuals with T2DM and obesity. Participants were randomised to either the lifestyle intervention ($n = 1\,918$) or to a control arm ($n = 1\,894$) that received health education only. The review reported prevalence rates for dementia in the intervention and control groups as 1.8% and 1.8% respectively. It further reported prevalence rates for MCI in the intervention and control group as 6.4% and 6.6% respectively. In completing the GRADE tables for the present review, the primary article (54) was utilised to ensure accurate event rates for both outcomes. These were then used to calculate an RR to be transformed into absolute estimates of effect, which overall indicated no meaningful difference in incident MCI or incident dementia at approximately 11.4 year follow up.

AMSTAR-II quality rating of this review: critically low

GRADE interpretation for the outcome of incident dementia

Due to the very low certainty evidence, we are unable to determine whether intensive diet and exercise intervention impacts in incident dementia in adults with T2DM and healthy cognitive function compared to controls, at a follow up of 11.4 years.

GRADE interpretation for the **outcome of incident MCI**

Due to the very low certainty evidence, we are unable to determine whether intensive diet and exercise intervention impacts in incident MCI in adults with T2DM and healthy cognitive function compared to controls, at a follow up of 11.4 years.

2. Sun et al. (22): Exercise

Sun et al. (22) completed a systematic review of the literature to synthesise the effect of multi-component exercise on cognitive function in individuals with T2DM. Searches were conducted across four databases from inception to December 2023. Twelve articles (55-66) (n = 1193) were included in the systematic review and meta-analysis, representing a variety of interventions including aerobic, resistance and multi-modal exercise. Included studies were deemed to be at a low to moderate risk of bias. Pooled analyses indicated exercise may have a large benefit for cognitive function (SMD = 0.91, 95% CI [0.28, 1.54]).

AMSTAR-II quality rating of this review: critically low

GRADE interpretation for the **outcome of cognitive function**

Due to the very low certainty evidence, we are unable to determine whether exercise intervention impacts cognitive function in adults with T2DM and healthy cognitive function compared to controls, at a follow up of 12 weeks to 96 weeks.

3. Zhang et al. (23): Combined aerobic and resistance exercise

A systematic review and meta-analysis conducted by Zhang et al. (23) explored the impact of combined aerobic and resistance exercise on health parameters such as cognition, metabolic health, and quality of life. Searches were conducted in 8 databases from inception to June 2023. RCTs conducted with middle-aged or older T2DM patients were included in this review. Sixteen RCTs were included in the overall review. A meta-analysis was conducted on five (67-71) of the included RCTs (n = 867) to explore both physical and mental Quality of Life assessed using the Short Form 36 (SF-36). The review reported that pooled analysis of the physical domains favoured combined aerobic and resistance training (n = 415) on global cognition (MD = 4.17, 95% [0.86, 7.48]) compared to control (n = 412). A similar effect was observed for the mental quality of life score (MD = 1.74, 95% CI [-3.46, 6.94]), however the review reported this result as not significant.

AMSTAR-II quality rating of this review: critically low

GRADE interpretation for the **outcome of cognitive function**

Due to the very low certainty evidence, we are unable to determine whether combined aerobic and resistance exercise impacts quality of life compared to usual care / placebo / no intervention in adults with healthy cognition at a follow up period of 12 weeks to 12 months.

3.3.4 Lifestyle intervention in T2DM patients with MCI

Population: Adults aged 40+ with T2DM and MCI

Intervention: Lifestyle intervention (exercise)

Comparator: Usual care / placebo / no intervention

Table WA.E.4d GRADE Summary of findings- lifestyle/diet interventions vs usual care/placebo/no intervention in adults with T2DM and MCI

Outcome	Intervention	Review results & measurement	Absolute effect estimates		Certainty of evidence	Summary
			Intervention	Comparator		
Cognitive Function	Exercise	Based on data from 594 participants in 6 RCTs ¹ Follow up: range 3 months to 12 months Assessed with: MMSE, MoCA	299 SMD = 0.65 95% CI [0.48, 0.82]	295	Moderate Due to serious risk of bias ^{a, b}	Exercise-based intervention likely benefits cognitive function in individuals with MCI and T2DM.

^a**Risk of Bias:** serious. All included studies are at an unclear risk of bias; the nature of the limitations may lower confidence in the estimates of effect.

^bThe review team acknowledges that the meta-analytic sample includes participants both with MCI and Healthy Cognition. Participants with MCI appear to constitute more than 50% of the sample. Evidence has not been downgraded for indirectness, as direct PICO match on all other components, and the population indirectness is unlikely to affect the overall quality of evidence.

References:

¹ Lu HH, Zhou Y, Chen C, Gu ZJ. Meta-analysis of the effect of exercise intervention on cognitive function in elderly patients with type 2 diabetes mellitus. BMC Geriatr. 2024;24(1):770.

Narrative Summaries

1) *Lu et al. (24): Exercise intervention*

There was minimal evidence identified for the use of diet/lifestyle interventions to reduce risk of dementia and cognitive decline in individuals with T2DM and MCI. One systematic review contributed towards this GRADE Table.

A systematic review and meta-analysis conducted by Lu et al. (24) synthesised available RCTs of any exercise intervention (including aerobic, resistance training, and balance training) on the Global Cognitive Function of T2DM patients aged 60+. Articles were included if they compared an exercise intervention to routine care or no intervention and measured cognitive function via the MoCA or MMSE. Search of eight databases from 1974 to July 2024 for articles published in English or Chinese, led to eight RCTs included in the review. Risk of Bias in the included studies was found to be low to unclear.

A meta-analysis was conducted of six of these RCTs (63, 65, 72-75), which suggested greater global cognitive function in the exercise (n = 299) compared to the control group (n = 295) (SMD = 0.65; 95% CI [0.48, 0.82]). This result was robust to measurement tool (i.e. MoCA, MMSE), intervention length, and intervention format (i.e. single, multi-modal). It should be noted however that the meta-analysis represented participants both with healthy cognition and MCI, and the review did not further stratify the sample. Consideration of the reference list indicates participants with MCI represented more than half of the meta-analytic sample.

AMSTAR-II quality rating of this review: critically low

GRADE interpretation for the **outcome of cognitive function**

The direction of effect indicates the treatment of T2DM using exercise likely has a benefit for cognitive function in adults with T2DM and MCI, compared to no intervention / usual care. The evidence suggests moderate certainty that exercise-based intervention treatment likely has a benefit for cognitive function at a follow up of 3 months to 12 months.

3.4 Additional evidence not reported in GRADE evidence profiles

This section reports additional relevant information presented in the systematic reviews included in this umbrella review, that was not reported in the GRADE tables. It should be noted that in line with the umbrella review methodology, we did not extract additional potentially relevant information from primary articles unless referred to in the included systematic review.

3.4.1 GLP-1-RAs and cognitive function in cognitively normal T2DM patients

In the review conducted by Garcia-Casares et al. (16), a second relevant RCT was identified but not included in the GRADE tables due to lacking any statistical comparison between the intervention and control groups. The RCT was conducted by Vadini et al. (76) and compared the administration of Liraglutide (n = 16) to lifestyle counselling (n = 16) in people with T2DM or intermediate hyperglycaemia, with a median follow up of 4 months. The article reported minimal change in composite memory for the control group from pre (MD = -0.48) to post-test (MD = -0.45) (p = 0.901), with a small effect size (d = 0.03). The intervention group however demonstrated a significant improvement from pre (MD = -0.68) to post-test (MD = 0.032) (p = 0.006), with a large effect size (d = 0.90). This trial was at an unclear risk of bias.

3.4.2 Traditional Chinese medicine for improving cognitive function

Su et al. (29) conducted a systematic review and meta-analysis to explore the efficacy and safety of combining Traditional Chinese and Western Medicine for patients with Diabetes Associated Cognitive Decline (DACD). Eight databases were comprehensively searched from inception to July 2023. Twenty RCTs were ultimately included in the review, with participants having a mean age ranging from 59 to 80. In the included studies, the intervention groups were administered Traditional Chinese Medicine in combination with glucose-lowering medication. The comparators in the included studies were glucose-lowering medication alone.

Twelve studies included in the review (n = 1 051) measured cognitive function through the MoCA. Pooled meta-analyses indicated potential improvements in MoCA scores for the intervention (n = 528) compared to the control (n = 524) groups (MD = 2.52, 95% CI [1.75, 3.30]). A similar effect was observed for cognitive function measured through the MMSE (MD = 2.31, 95% CI [1.33, 3.29]) in 11 studies (n = 802; 401 / 401). These results suggest that combined Traditional Chinese Medicine and glucose-lowering

medication may have a better ability to preserve cognitive function in individuals with DACD, than standard glucose-lowering medication alone.

Further, 12 of the included studies reported on adverse reactions, with 5 affirming them to have occurred. Meta-analyses indicated differences in Adverse Reactions did not substantially vary between groups (OR = 0.40; 95% CI [0.12, 1.31]). This suggests the safety of administering Traditional Chinese Medicine in combination with standard hypoglycaemic medications is comparable to hypoglycaemic medication alone.

3.4.3 Adverse events

Two reviews identified by our search but not included in the GRADE evidence profiles reported on adverse events. This information was not included in the GRADE tables due to the absence of substantial quantitative information provided at the review level. Tang et al. (18) narratively reported on adverse events in 3 RCTs that compared intensive versus standard glucose control in T2DM patients with healthy cognition. It was indicated that mortality in two of the relevant RCTs was higher for participants assigned to receive intensive glucose control than those assigned to conventional. The third relevant RCT had the converse finding, however.

Cai et al.. (26) conducted a systematic review and meta-analysis evaluating the impact of physical activity intervention on cognitive function in T2DM patients aged 60+. The review included five RCTs (n = 738), 2 of which (60, 65)(n = 450) comprised participants with healthy cognition at baseline. Included studies represented various exercise intervention modalities, including aerobic training, resistance training, and multi-modal exercise. Cai et al. narratively reported no adverse events in the two trials of exercise-based intervention on cognitively normal T2DM patients.

3.4.4 Drop-out rates

One review identified by our search but not included in the GRADE evidence profiles reported on drop-out rates. Jin et al. (25) conducted a systematic review and meta-analysis which evaluated the relative effects of GLP1-RAs and DPP-4i on cognitive function in individuals with T2DM. Two of the included studies (41, 77) were RCTs comprised of participants with healthy cognition at baseline and drop-out rates were reported narratively in these studies. The first RCT conducted by Biessels et al. (41) reported a 0% loss to follow up in 1545 participants. The second, conducted by Perna et al. (77) reported a 10% loss to follow up in 39 participants. No further information on between group differences was provided in the review.

3.4.5 Systematic reviews, meta-analyses and network analyses of primarily observational data

As part of the methodology for the current evidence profile, the eligibility criteria were restricted to include only systematic reviews of RCTs. This is because RCTs provide the highest quality evidence of any primary research design due to their rigorous procedures enabling strong causal inference. Following this process however did not yield any intervention evidence on several key first and second-line glucose-lowering medicines (e.g., metformin, sulfonylureas), that are commonly prescribed as per clinical guidelines (10). Therefore, two recent systematic reviews based on primarily observational data

have been identified through expert knowledge of the field and are reported below to provide further context and best inform evidence to decision making, Please note, this additional information's role in the present evidence profile should be understood as selective, and not subject to the methodology of this umbrella review. As such, it may not represent the full evidence base and should be considered in supplement only.

Tian et al. (78): Glucose-lowering medications and cognitive outcomes

A systematic review and network meta-analysis was conducted by Tian et al. (78) to summarise the effect of glucose-lowering medications/ anti-diabetes agents on cognitive outcomes. Twenty-seven studies were included in the review, comprising 3 RCTs, 19 cohort studies, and 5 case-control studies. The total number of participants represented was 1 590 757. The following is a summary of the results:

- Glucose-lowering medication compared to no treatment on risk of dementia:
 - SGLT-2 Inhibitors: OR 0.41 (95% CI, 0.22–0.76).
 - GLP-1 Receptor Agonists: OR 0.34 (95% CI, 0.14–0.85).
 - Thiazolidinedione: OR 0.60 (95% CI, 0.51–0.69).
 - DPP-4 Inhibitors: OR 0.78 (95% CI, 0.61–0.99).
 - Sulfonylurea: OR 1.43 (95% CI, 1.11–1.82) - increased risk.
- Network Meta-Analysis Rankings:
 - SGLT-2 Inhibitors: Best (SUCRA = 94.4%).
 - GLP-1 Receptor Agonists: Second best (SUCRA = 92.7%).
 - Thiazolidinedione: Third best (SUCRA = 74.7%).
 - DPP-4 Inhibitors: Fourth best (SUCRA = 54.9%).
 - Sulfonylurea: Second worst (SUCRA = 20.0%).
- Effectiveness for Delaying Cognitive Impairment, Dementia, and AD:
 - SGLT-2 Inhibitors $\approx$ GLP-1 Receptor Agonists > Thiazolidinedione > DPP-4 Inhibitors.
 - Sulfonylurea: Highest risk.

Conclusion: Provides evidence for evaluating treatment options in clinical practice.

Zhou et al. (79): Glucose-lowering medications and cognitive outcomes

Zhou et al. (79) conducted a network meta-analysis (9 studies, 530 355 individuals) and meta-analysis (17 studies, 1 258 879 individuals) to evaluate the impact of different glucose-lowering medications on risk of dementia in T2DM individuals. GLP-1 receptor agonists and SGLT-2 inhibitors were excluded due to lack of data. The following is a summary of results:

- Glucose-lowering medication compared to no treatment on risk of dementia:
 - DPP-4 Inhibitors: HR 0.54 (95% CI, 0.38–0.74) also for AD risk HR 0.48 (95% CI, 0.25–0.92).
 - Metformin: HR 0.75 (95% CI, 0.63–0.86).
 - Sulfonylurea: HR 0.85 (95% CI, 0.73–0.98).
 - Thiazolidinedione: HR 0.70 (95% CI, 0.55–0.89).
 - Node-splitting analysis should inconsistency of direct and indirect estimates for sulfonylurea (P=0.042).
- Following agents exhibit benefit over insulin for dementia risk reduction:
 - DPP-4 Inhibitors: HR 0.35 (95% CI, 0.20–0.59).
 - Metformin: HR 0.48 (95% CI, 0.30–0.77).

- Thiazolidinedione: HR 0.45 (95% CI, 0.29–0.73).
- Sulfonylurea: HR 0.55 (95% CI, 0.34–0.88).
- Meta-Analysis Findings:
 - Metformin: HR 0.86 (95% CI, 0.74–1.00).
 - DPP-4 Inhibitors: HR 0.65 (95% CI, 0.55–0.76).
 - Insulin: Increased risk of AD: HR 1.60 (95% CI, 1.13–2.26).
 - GLP-1 Analogs and SGLT-2 Inhibitors: No significant association with dementia from two case-control studies (GLP-1 analogs: OR 0.71; 95% CI, 0.46–1.10 and SGLT-2 inhibitors: OR 0.74; 95% CI, 0.47–1.15).

Conclusion:

- **Lowest Risk:** DPP-4 inhibitors, **followed by:** metformin and thiazolidinedione.
- **Highest Risk:** Insulin.
- **Future Research:** Needed for GLP-1 analogs and SGLT-2 inhibitors.

World Health Organization (15): Comparison of glucose-lowering medications in the treatment of T2DM

Table WA.E.5 Hypoglycaemic agents for third-line treatment in type 2 diabetes

HbA1c (mean difference in change from baseline with 95% confidence interval)

TZD	0.23 ⊕○○○ ^{b, c} (-0.62, 1.08)	0.12 ⊕○○○ ^{b, c} (-1.12, 1.35)	-0.00 ⊕○○○ ^{b, c} (-0.61, 0.61)	0.86 ⊕○○○ ^{a, c} (0.25, 1.48)
DPP-4 inhibitors		-0.12 ⊕○○○ ^{b, c} (-1.52, 1.29)	-0.23 ⊕○○○ ^{b, c} (-1.03, 0.56)	0.63 ⊕○○○ ^{b, c} (-0.29, 1.55)
		SGLT-2 inhibitors	-0.12 ⊕○○○ ^{b, c} (-1.39, 1.15)	0.75 ⊕○○○ ^{b, c} (-0.32, 1.82)
		Basal insulin		0.86 ⊕○○○ ^{a, c} (0.18, 1.55)
				Placebo

^a Study limitations/risk of bias

^b Imprecision

^c Inconsistency (heterogeneity and incoherence)

3.5 Research gaps

Interventions:

The present review has highlighted a substantial lack of evidence considering the impact of pharmacological or diet/lifestyle intervention on cognitive outcomes specifically in populations with MCI. This is a critical gap in the literature, as a lack of understanding of optimal/effective time for intervention as it relates to cognition status, undermines the ability to make accurate and effective recommendations in practice.

Lack of pooled evidence for differential effects of glucose-lowering medicines, including comparisons with WHO recommended first- and second-line treatments for T2DM (i.e. metformin, second-generation sulfonylurea (preferably gliclazide) and insulin). Observational data suggest that certain agents may be

more effective in reducing dementia risk or risk of cognitive decline (i.e. SGLT-2 Inhibitors $\approx$ GLP-1 Receptor Agonists > Thiazolidinedione > DPP-4 Inhibitors), but this needs to be confirmed by RCT evidence.

There was further scarce evidence for lifestyle-based interventions beyond those that are exercise-based. Broader types of lifestyle interventions (e.g., diet, weight loss) may be critical in T2DM management, and have important implications for cognitive function.

Outcomes and populations:

Most of the identified evidence pertains to cognitive function. Whilst this is an important outcome to consider in the context of preventing dementia and cognitive decline, further research is required to discern whether this evidence translates to meaningful differences in Incident Dementia or Incident MCI. Similarly, patient important outcomes such as quality of life and functional status were largely unreported in the included reviews, and should be considered in tandem to support clinical decision making.

Finally, future research would benefit from considering potentially important subgroups such as age, gender, and ethnicity, to ensure results are translatable across these groups. The present evidence largely represents higher income countries, and a broader representation is needed, particularly given the differences observed in both T2DM and dementia prevalence rates in lower/middle income countries compared to these higher ones.

The overall evidence profile has not substantially changed since the previous guidelines, with the ability to make appropriate recommendations limited by the very low to moderate certainty of evidence.

4. From evidence to recommendations

4.1 Evidence-to-decision table

Table WA.E.5 Evidence-to-decision table

Is the problem a priority?	
Judgement	Research evidence and additional considerations
Yes	In 2021, there were 57 million people globally living with dementia with over 60% in low-and middle-income countries(1). The number of people living with dementia is projected to increase to 153 million by 2050 (2). Dementia is one of the major causes of disability and dependency among older people worldwide (1). As there is still no cure for dementia, risk reduction for cognitive decline and dementia is important. T2DM has been identified as one of the key risk factors for dementia (8), In 2021, there were 536.6 million people worldwide affected by diabetes (80), approximately 90% by T2DM. Diabetes is the 8th leading cause of death and disability globally (81) and attributed to \$966 billion (USD) in health expenditure around the world (80). Importantly, T2DM is generally considered modifiable by pharmacological and/or lifestyle intervention (3). Given its high prevalence, and as a potential risk factor for dementia and status as a critical public health priority itself, T2DM may be a pivotal target for reducing cognitive decline and dementia. Diabetes is a chronic, metabolic disease characterized by elevated levels of blood glucose (or blood sugar), which leads over time to serious damage to the heart, blood vessels, eyes, kidneys and nerves. For people living with diabetes, access to affordable treatment is vital for preventing mortality and reducing the risk of complications. However, in 2022, more than half (59%) of adults aged 30 years and over living with diabetes were not taking medication for their diabetes. Diabetes treatment coverage was lowest in low- and middle-income countries (7).
How substantial are the desirable anticipated effects?	
Judgement	Research evidence and additional considerations
Small	Cognitive outcomes in adults with T2DM and normal cognition Importantly, no intervention evidence was identified for WHO recommended first and second -line pharmacological treatments of T2DM (i.e. metformin and sulfonylurea) and their effects on cognitive decline and dementia/MCI incidence (10, 15)

The reported evidence suggests **small benefits on cognitive function outcomes for DPP-4i and GLP-1 RA** and no meaningful effects for intensive glucose control vs standard treatment (for either dementia incidence or cognitive function measures). Lifestyle interventions, especially **physical exercise/activity may have large benefits for cognitive outcomes and quality of life**. However, the evidence for all outcomes is of very low to low certainty.

The search also identified very low-quality evidence for the use of intranasal insulin with potential small benefits on cognitive outcome measures. While not a treatment for T2DM, evidence was considered due to specific relevance of intranasal insulin for dementia prevention and for the sake of full transparency. However, due to very low certainty of the evidence no meaningful conclusion can be drawn.

Cognitive outcomes in adults with T2DM and MCI

Identified evidence further suggests large benefits on cognitive function for DPP-4i and physical exercise/activity in people with T2DM and MCI, with very low and moderate certainty, respectively. However, it should be noted that this judgement is based on a sample which was confounded with cognitively normal participants.

Evidence gaps

There was no evidence in the present review of a desirable effect of lifestyle or pharmacological treatment of T2DM to reduce risk of **incident Dementia or incident MCI**. It is important to note that this is largely due to these binary outcomes not being measured and/or based on single trial evidence. As such, the evidence remains uncertain.

Gradable intervention evidence for common treatments for T2DM such as metformin, sulfonylureas, sodium-glucose co-transporters type 2 (SGLT-2) inhibitors or insulin injections on reducing risk for cognitive decline and dementia has not been identified. Similarly, no gradable intervention evidence has been identified for the superiority of one hypoglycaemic agent over another for their potential to reduce or prevent cognitive decline/dementia. Observational evidence suggests that SGLT-2 inhibitors (similarly to GLP-1 RA) are associated with the lowest dementia risk, followed by TZD, followed by DPP-4i. Sulfonylurea carried the highest risk for dementia in included network meta-analyses.

One of the most important ways to treat diabetes is to maintain a healthy lifestyle. In addition, most people with T2DM will require glucose-lowering medicines (7).

How substantial are the undesirable anticipated effects?

Judgement	Research evidence and additional considerations
Small	The included systematic reviews provided no certain evidence regarding potential for undesirable effects. It is difficult to make a confident judgement as review-level reporting on signifiers of undesirable effects (e.g., adverse events) was largely absent. Where they were reported, insufficient detail was provided to understand the nature of adverse events or drop-outs and thus draw a meaningful conclusion. There was no certain evidence that any of the evaluated interventions had a negative impact on considered outcomes, however some confidence intervals did overlap. Of note, hypoglycaemia is an adverse effect for certain glucose-lowering therapies, particularly insulin and sulfonylureas (82). Hypoglycaemia, particularly when severe or recurrent, has been associated with an increased risk of cognitive decline and dementia in people with T2D (82). This underscores the importance of avoiding overly intensive glycaemic control in older adults and those at risk of severe hypoglycaemia (10)

What is the overall certainty of evidence?

Judgement	Research evidence and additional considerations
Very Low	There is overall a very low certainty of evidence of effects. Where low to moderate certainty and promising evidence exists, there remains concerns about the small number of trials contributing to the evidence, and whether the observed change is clinically meaningful. The certainty of evidence for management of diabetes varies for different interventions/ glucose-lowering medicines from very low to moderate but is generally considered effective (15)

Is there important uncertainty about or variability in how much people value the main outcomes?

Judgement	Research evidence and additional considerations
Probably no important uncertainty or variability	Curran et al 2021 (83) conducted a systematic review of qualitative evidence (all conducted in HIC) on what the general population thinks about dementia risk reduction. They narratively synthesised the results and stated that participants generally feared dementia and valued lower dementia risk. However, there were additional considerations such as the plausibility and effectiveness of risk reduction, short term outcomes such as quality of life and intervention burden which also played a part in decision making. Although research on the perception of dementia prevention in LMICs remains limited, this area of study is crucial because the majority of

people with dementia live in LMICs. Current implementation randomized control trial in Africa indicates that for dementia prevention interventions to be effective and valued, they must be tailored to local cultural contexts (84). Consistent public health messaging for both the general public and health professionals on dementia prevention is also essential, as highlighted by feedback from a consensus workshop on dementia prevention in Latin America (85).

The WHO Model List of Essential Medicines (86) for diabetes includes 9 medicines and 5 therapeutic equivalents for treating diabetes. In recent years, options for oral hypoglycaemic agents (OHAs) have increased. Aside from metformin and sulfonylureas, DPP4 inhibitors, SGLT-2 inhibitors, and TZDs are now available and recommended for treatment augmentation in various high-income settings. However, these **new OHAs are quite expensive and often cost more than human insulin**, which could be an alternative option for treatment augmentation.

When choosing a treatment, clinical goals need to be balanced with patient preferences and well-being.

Judgement in WHO diabetes guidelines for values probably favoured DPP-4 I, SGLT-2 inhibitors, or glitazones over insulin (10).

Does the balance between desirable and undesirable effects favour the intervention or the comparison?

Judgement

Research evidence and additional considerations

Probably favours the intervention

Overall, the balance between desirable and undesirable effects probably favours the intervention. In the available evidence, there is little to suggest undesirable effects related to the interventions. However, this may be due to lack of reporting at the review level. Whilst the effectiveness of each intervention in supporting cognition and reducing dementia risk varies, there is no or minimal evidence of the interventions negatively impacting on cognition or dementia risk compared to the controls.

As noted above the general population typically reports feelings of fear regarding dementia, and see value in lowering dementia risk (83). While recognising a tendency to prioritise short-term (e.g., a focus on quality of life), this population does not appear to overall value these effects more but rather consider them in complement to their overall health (of which dementia risk forms part of).

Hypoglycaemia is a frequent iatrogenic complication in diabetic patients, occurring particularly in patients receiving sulfonylurea or insulin. Hypoglycaemia, risk should be minimised. Hypoglycaemia can be managed (See *HEARTS D: diagnosis and management of type 2 diabetes* (10) for prevention and management of diabetes complications).

Given the serious health risks if diabetes is untreated, the benefits outweigh risks for overall health.

How large are the resource requirements (costs)?	
Judgement	Research evidence and additional considerations
Negligible costs and savings	No data on costs was reported in the included reviews. The costs may vary largely. Pharmacological approaches will predominantly accrue costs for the medication, administration tools, and continuous prescription and monitoring needs. Diet/lifestyle approaches will vary even further depending on the nature of the intervention, but could include exercise access fees (e.g., gym/class membership), professional support fees (e.g., dietitian, personal trainer), food expenses, etc. Diabetes management carries no additional cost for dementia prevention, as treatment costs arise solely from its primary clinical indication. The WHO recommended first-line treatment for T2DM is metformin as it does not cause weight gain or hypoglycaemia and is the recommended initial treatment for people who do not achieve the desired glycaemic control with diet and physical activity. A second-generation sulfonylurea (preferably gliclazide) can be used as initial (first line) treatment when metformin is contraindicated or not tolerated. However, sulfonylureas may cause weight gain or hypoglycaemia. Other OHAs such as DPP-4i, SGLT-2 inhibitors or thiazolidinedione (TZD) may be added if metformin, sulfonylureas or insulin fail to achieve glucose control, but generally not recommended for routine use due to high cost and, with the exception of SGLT-2 inhibitors, uncertain benefit (10)
What is the certainty of the evidence of resource requirements (costs)?	
Judgement	Research evidence and additional considerations
No included studies	No data on the certainty of costs was reported in the included reviews.

Does the cost-effectiveness of the intervention favour the intervention or the comparison?	
Judgement	Research evidence and additional considerations
No included studies	No data on the cost-effectiveness of glucose-lowering agents or diet/lifestyle intervention was reported in the included reviews. According to WHO, a series of cost-effective interventions can improve patient outcomes, regardless of type of diabetes. These interventions include blood glucose control through a combination of diet, physical activity and, if necessary, medication; control of blood pressure and lipids to reduce cardiovascular risk and other complications; and regular screening for damage to the eyes, kidneys and feet to facilitate early treatment (7).
What would be the impact on health equity, equality and non-discrimination?	
Judgement	Research evidence and additional considerations
Probably increased	Most studies on dementia risk reduction are conducted in high income countries (87). A number of studies have shown that people with lower education and low SES are at higher risk of developing dementia (88, 89). Additionally, several risk factors are more prevalent in LMIC and/or among specific ethnicities (e.g. hypertension, diabetes etc (90)). While there are cost and implementation considerations that vary between interventions (e.g. the cost of hearing aids, cataract surgery, depression medications etc). Overall treatment of risk factors for dementia is anticipated to increase health equity. The majority of 830 million people with diabetes live in low-and middle-income countries. More than half of people living with diabetes are not receiving treatment and both the number of people living with diabetes and not having their diabetes treated has been increasing steadily over the past decades.(91) The affordability and accessibility of interventions likely vary between the noted interventions and the types of approaches that may be taken to implement them.

The oral hypoglycaemic agents studied in the context of dementia risk reduction are generally more expensive and access in low-resource settings may be limited.

Due to the immense global burden of diabetes and the large treatment gaps observed globally despite effective and cost-effective treatments being available, successful implementation of these interventions would be anticipated to increase health equity and improve health outcomes.

Is the intervention feasible to implement?

Judgement	Research evidence and additional considerations
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Probably yes	As treatment of T2DM is already integrated into clinical practice and recommendations, doing so for the purposes of reducing dementia risk should generally be feasible. However, differences in access to resources around the world (especially in LMIC) as well as potential attitude differences across cultures may present barriers to implementation.
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Is the intervention aligned with human rights principles and socio-culturally acceptable?

Judgement	Research evidence and additional considerations
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Yes	No data on sociocultural acceptability of hypoglycaemic agents or diet/lifestyle intervention was reported in the included reviews.
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The benefits of pharmacological and nonpharmacological treatment and management of T2DM are well-established and strongly recommended for their overall health benefits, indicating that treatment of diabetes in general is in line with the basic right to health.

Acceptability of or preferences for different OHAs may vary by drug class but may also be heavily influenced by availability and costs.

5. Summary of judgments

Table WA.E.6 Summary of judgements

Criterion	Judgement						
Priority of the problem	Don't know	Varies	---	No	Probably no	Probably yes	Yes
Desirable effects	Don't know	Varies	---	Trivial	Small	Moderate	Large
Undesirable effects	Don't know	Varies	---	Large	Moderate	Small	Trivial
Certainty of the evidence		---	---	Very low	Low	Moderate	High
Values	---	---	---	Important uncertainty or variability	Possibly important uncertainty or variability	Probably no important uncertainty or variability	No important uncertainty or variability
Balance of effects	Don't know	Varies	Favours comparison	Probably favours comparison	Does not favour either	Probably favours intervention	Favours intervention
Resources required	Don't know	Varies	Large costs	Moderate costs	Negligible costs and savings	Moderate savings	Large savings
Certainty of the evidence on required resources	No included studies	---	---	Very low	Low	Moderate	High
Cost-effectiveness	No included studies	Varies	Favours comparison	Probably favours comparison	Does not favour either	Probably favours intervention	Favours intervention
Health equity, equality and non-discrimination	Don't know	Varies	Reduced	Probably reduced	Probably no impact	Probably increased	Increased
Feasibility	Don't know	Varies	---	No	Probably no	Probably yes	Yes
Human rights and socio-cultural acceptability	Don't know	Varies	---	No	Probably no	Probably yes	Yes

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Appendix 1 – Search strategy

Embase Search (Elsevier)	Results
1 ((lower* OR reduc* OR loss OR losses OR slower OR better OR improv* OR effect OR Effects OR efficac* OR affect* OR impact* OR persist* OR further OR detect* OR change OR vary OR changing OR risk OR ratio* OR Odds OR impact OR detriment* OR influenc* OR maintenance OR protect* OR enhanc* OR prevent* OR preserv* OR greater OR restor* OR progress* OR adverse OR beneficial OR performance* OR score* OR compensat*) NEAR/8 (cognitive-measurement* OR cognitive-enhanc* OR Cognitive-improv* OR cognitive-outcome* OR cognitive-domain* OR cognitive-skill* OR cognitive-change* OR cognitive-performanc* OR neurocogniti* OR cognition* OR cognitive-function* OR cognitive-decline* OR cognitive-dysfunction* OR cognitive-impairment* OR cognitive-deficit* OR cognitive-behaviour* OR dementia* OR cognitive-abilit* OR alzheimer* OR cognitive-benefit* OR neurodegeneration OR neuro-degenerat* OR neuroprotective* OR MCI OR neurophysiologic* OR Neuropsycholog* OR psychosocial-function* OR psychosocial-outcome* OR executive-function* OR memory OR attention-deficit*)) :ti,ab	597 339
2 ('diabetes mellitus'/exp AND 'therapy'/exp) OR ((diabet* OR niddm OR (insulin NEAR/2 resistanc*)) AND (intervention* OR treatment* OR treated OR control OR program* OR therap* OR pharmaco* OR management OR diet* OR exercis* OR lifestyle* OR physical-activit*)) :ti,ab,kw,de OR 'antidiabetic agent'/exp OR 'dipeptidyl peptidase IV inhibitor'/exp OR '2,4 thiazolidinedione derivative'/exp OR 'glucagon like peptide 1 receptor agonist'/exp OR 'alpha glucosidase inhibitor'/exp OR ((antidiabetic* OR anti-diabetic OR hypoglyc*) NEAR/2 (agent* OR drug* OR compound* OR herb* OR sulfonamide* OR antidiabetic*)) :ti,ab OR (insulin OR metformin OR "DPP-4-inhibito*" OR dipeptidyl-peptidase-IV-inhibitor* OR gliptin* OR "SGLT2 inhibito*" OR "sodium-glucose-cotransporter-2-inhibitor*" OR gliflozin* OR sul*onylurea OR sul*onurea OR sul*onyl-urea OR sulfonylcarbamide* OR thiazolidinediones OR glitazone* OR	1 760 738

"GLP-1 agonists" OR "glucagon-like-peptide-1-receptor-agonist*" OR "alpha glucosidase inhibit*" OR meglitinide*):ti,ab

3	#1 AND #2	20 665
4	[animals]/lim NOT ([animals]/lim AND [humans]/lim)	6 602 945
5	[letter]/lim OR [conference abstract]/lim OR [conference paper]/lim OR [conference review]/lim OR [editorial]/lim OR [note]/lim OR 'preprint'/exp	9 358 756
6	#3 NOT (#4 OR #5)	12 363
7	'meta analysis'/exp OR 'systematic review'/exp OR (review* OR overview*):ti OR ((meta NEAR/3 analy* OR metaanaly* OR (synthes* NEAR/3 (literature* OR research* OR studies OR data)) OR (pooled NEAR/3 analys* OR ((data NEAR/2 pool*) AND studies) OR medline OR medlars OR embase OR cinahl OR scisearch OR psychinfo OR psycinfo OR psychlit OR psyclit OR cinhal OR cancerlit OR cochrane OR bids OR pubmed OR ovid OR ((hand OR manual OR database* OR computer*) NEAR/2 search*) OR (electronic NEAR/2 (database* OR 'data base' OR 'data bases'))):ti,ab OR bibliograph*:ab OR 'relevant journals':ab OR ((review* OR overview*) NEAR/10 (systematic* OR methodologic* OR quantitativ* OR research* OR literature* OR studies OR trial* OR effective*)):ab	2 109 714
8	#6 AND #7	2 467
9	#8 AND [2020–2025]/py	1 504

MedLine (Ovid)	Search	Results
1	((lower* OR reduce* OR loss OR losses OR slower OR better OR improv* OR effect OR effects OR efficac* OR affect* OR impact* OR persist* OR further OR detect* OR change OR vary OR changing OR risk OR ratio* OR odds OR impact OR detriment* OR influenc* OR maintenance OR protect OR enhanc* OR prevent* OR preserv* OR greater OR restor* OR progress* OR adverse OR beneficial OR performance* OR score* OR compensat*) adj9 ((cognitive adj (measurement* OR enhanc* OR improv* OR outcome* OR domain* OR skill* OR change* OR performanc* OR function* OR decline* OR dysfunction* OR impairment* OR deficit* OR behaviour* OR behavior* OR abilit* OR benefit*)) OR neurocogniti* OR cognition* OR dementia* OR alzheimer* OR neurodegeneration OR neuro-degenerat* OR neuroprotective* OR MCI OR neurophysiologic* OR Neuropsycholog* OR (psychosocial adj (function* or outcome*)) OR executive function* OR memory OR attention deficit*)):ti,ab.	466 159

2	(exp Diabetes Mellitus/ AND exp Therapeutics/) OR ((diabet* OR niddm OR (insulin adj2 resistanc*)) AND (intervention* OR treatment* OR treated OR control OR program* OR therap* OR pharmaco* OR management OR diet* OR exercis* OR lifestyle* OR physical-activit*)).ti,ab,kw,de OR exp Hypoglycemic Agents/ OR ((antidiabetic* OR anti-diabetic OR hypoglyc*) adj2 (agent* OR drug* OR compound* OR herb* OR sulfonamide* OR antidiabetic*)).ti,ab OR (insulin OR metformin OR "DPP-4-inhibito*" OR dipeptidyl-peptidase-IV-inhibitor* OR gliptin* OR "SGLT2 inhibito*" OR "sodium-glucose-cotransporter-2-inhibitor*" OR gliflozin* OR sul*onylurea OR sul*onurea OR sul*onyl-urea OR sulfonylcarbamide* OR thiazolidinediones OR glitazone* OR "GLP-1 agonists" OR "glucagon-like-peptide-1-receptor-agonist*" OR "alpha glucosidase inhibit*" OR meglitinide*).ti,ab	960 478
3	1 AND 2	16 208
4	animals/ not (animals/ and humans/)	5 257 137
5	(letter or historical article or comment or editorial or news or case reports).pt.	5 074 320
6	3 NOT (4 OR 5)	12 138
7	((Meta-Analysis as Topic/ or meta analy\$.tw. or metaanaly\$.tw. or Meta-Analysis/ or (systematic adj (review\$1 or overview\$1)).tw. or exp Review Literature as Topic/ or cochrane.ab. or embase.ab. or (psychlit or psyclit).ab. or (psychinfo or psycinfo).ab. or (cinahl or cinhal).ab. or science citation index.ab. or bids.ab. or cancerlit.ab. or reference list\$.ab. or bibliograph\$.ab. or hand-search\$.ab. or relevant journals.ab. or manual search\$.ab.) or ((selection criteria.ab. or data extraction.ab.) and review/))	642 555
8	6 AND 7	720
9	limit 8 to yr=2020—2024	388
Cochrane Search Library		Results
1	((((lower* OR reduce* OR loss OR losses OR slower OR better OR improv* OR effect OR effects OR efficac* OR affect* OR impact* OR persist* OR further OR detect* OR change OR vary OR changing OR risk OR ratio* OR odds OR impact OR detriment* OR influenc* OR maintenance OR protect OR	282 475

enhanc* OR prevent* OR preserv* OR greater OR restor* OR
progress* OR adverse OR beneficial OR performance* OR
score* OR compensat*) NEAR/9 ((cognitive NEXT
(measurement* OR enhanc* OR improv* OR outcome* OR
domain* OR skill* OR change* OR performanc* OR function*
OR decline* OR dysfunction* OR impairment* OR deficit* OR
behaviour* OR behavior* OR abilit* OR benefit*)) OR
neurocogniti* OR cognition* OR dementia* OR alzheimer* OR
neurodegeneration OR neuro-degenerat* OR
neuroprotective* OR MCI OR neurophysiologic* OR
Neuropsycholog* OR (psychosocial NEXT (function* or
outcome*)) OR executive function* OR memory OR attention
deficit*)))):ti,ab,kw

2	MeSH descriptor: [Diabetes Mellitus] explode all trees	47 384
3	MeSH descriptor: [Therapeutics] explode all trees	439 359
4	#2 AND #3	15 723
5	((((diabet* or niddm or (insulin NEAR/2 resistanc*)) and (intervention* or treatment* or treated or control or program* or therap* or pharmaco* or management or diet* or exercis* or lifestyle* or physical NEXT activit*)))):ti,ab,kw	129 624
6	MeSH descriptor: [Hypoglycemic Agents] explode all trees	12 713
7	((((antidiabetic* or "anti diabetic" or hypoglyc*) NEAR/2 (agent* or drug* or compound* or herb* or sulfonamide* or antidiabetic*)))):ti,ab,kw	17 147
8	((insulin or metformin or DPP NEXT 4 NEXT inhibito* or dipeptidyl NEXT peptidase NEXT IV NEXT inhibitor* or gliptin* or SGLT2 NEXT inhibito* or sodium NEXT glucose NEXT cotransporter NEXT 2 NEXT inhibitor* or gliflozin* or sul*onylurea or sul*onurea or sul*only NEXT urea or sulfonylcarbamide* or thiazolidinediones or glitazone* or GLP NEXT 1 NEXT agonists or glucagon NEXT like NEXT peptide NEXT 1 NEXT receptor NEXT agonist* or alpha NEXT glucosidase NEXT inhibit* or meglitinide*)):ti,ab,kw	87 772
9	#4 OR #5 OR #6 OR #7 OR #8	153 755
10	#1 and #9	39

Limits Applied to Line 10:
Cochrane Reviews;
Jan 2020 to Dec 2024.

Appendix 2 –Data extraction template

Reference

LastName et al., (Year)

Notes for Selection

Provide details of all potential PICO matches in a review, completing the template for any additional intervention/outcome combinations. This section should be filled out based on what is available within the analyses of the review, as opposed to the review's eligibility criteria alone. Utilize the key below to identify the clarity of each match:

AMSTAR-II rating: _____

Overall Quality RANKING: _____

Potential PICO Match/es:

- Population (Age, Diabetes Status, Cognition Status):
- Intervention (Lifestyle/Pharmacological; Specific Intervention):
- Comparator:
- Outcome (Outcome Type; Synthesis Type):
- No. Included RCTs

Key:

Exact PICO Match

Close PICO Match

Non-PICO Match

Details to support selection for GRADE

Type of intervention

Type of synthesis for cognitive outcomes

Type of included studies for cognitive outcomes

Cognition Status at Baseline?

Relevant Outcomes of Interest:

- Cognitive Function
- Incident MCI
- Incident Dementia
- Quality of Life
- Functional Level
- Adverse Events
- Drop-Out Rates

Details for synthesis

Aim of systematic review

Basic search strategy

- Number of databases searched
- Search dates

Relevant Eligibility Criteria:

- Population?
 - Intervention?
 - Comparator?
 - Outcomes?
-

Characteristics of Included**Studies**

- *Number of Included RCTs*
 - *Sample Size*
 - *Measurement of Outcomes*
-

Risk of Bias in Included Studies**Overview of Results**

Appendix 3 – AMSTAR-II ratings

Reference	AMSTAR-II items																Non-critical weaknesses	Critical flaws (partial yes)	Critical flaws	Rating
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16				
Jin (2020)	Y	Y	Y	N	Y	Y	Y	PY	Y	N	Y	Y	Y	Y	Y	Y	1	0	1	Low
Su (2023)	Y	Y	Y	PY	N	Y	N	PY	Y	N	Y	N	Y	Y	Y	Y	3	1	1	Low
Sun (2024)	Y	Y	N	N	Y	Y	N	PY	Y	N	Y	Y	Y	Y	Y	Y	2	0	2	Critically low
Wu (2023)	Y	N	N	PY	Y	Y	N	PY	Y	N	Y	Y	Y	Y	Y	Y	2	1	2	Critically low
Nijssen (2023)	N	PY	N	N	Y	N	N	PY	Y	N	-	-	Y	N	-	Y	5	1	2	Critically low
Dyer (2020)	Y	N	N	N	Y	Y	N	Y	Y	N	-	-	Y	Y	-	Y	2	0	3	Critically low
Lu (2024)	Y	Y	Y	N	Y	Y	N	Y	Y	N	Y	N	N	Y	Y	Y	2	0	3	Critically low
Cooke (2020)	Y	N	Y	Y	N	N	N	Y	Y	N	Y	Y	Y	Y	N	Y	3	0	3	Critically low
Meng (2023)	Y	N	N	PY	Y	Y	N	N	Y	N	Y	N	N	Y	Y	Y	4	1	3	Critically low
Crabtree (2022)	Y	N	Y	N	Y	N	Y	PY	Y	N	N	Y	Y	N	N	Y	3	0	4	Critically low
Tang (2021)	Y	N	Y	N	N	Y	N	N	N	N	Y	Y	Y	Y	Y	Y	3	0	4	Critically low
Chai (2023)	Y	N	N	N	Y	Y	N	PY	Y	N	Y	N	N	N	Y	Y	4	0	4	Critically low

Yuan (2024)	Y	N	N	PY	Y	N	N	N	Y	N	Y	N	N	Y	N	Y	5	1	4	Critically low
Cai (2022)	Y	N	N	PY	Y	Y	N	Y	N	N	Y	N	N	Y	N	Y	3	1	5	Critically low
Zhang (2024)	Y	N	N	N	Y	Y	N	Y	Y	N	Y	N	N	Y	N	Y	3	0	5	Critically low
Ding (2024)	Y	N	N	N	Y	Y	N	PY	N	N	Y	N	N	N	Y	Y	4	0	5	Critically low
Garcia-Casares (2023)	N	N	N	N	N	N	N	PY	N	N	-	-	N	Y	-	Y	5	0	5	Critically low
Slouha (2023)	N	N	N	N	N	N	N	N	N	N	-	-	N	Y	-	Y	6	0	5	Critically low

Key:

Y Yes

PY Partial Yes

N No

Appendix 4 – Characteristics of individual studies included in the GRADE profiles

Note: Details about primary studies are presented as reported in the systematic reviews included in the present evidence profiles. Data (e.g., sample size) was extracted in the first instance from an available meta-analysis, then further from the characteristics of included studies or broader systematic review. The primary articles were only referred to when the review had either unclear reporting or did not report one or more characteristics.

Table WA.E.4a GRADE Summary of findings- Glucose-lowering medicines/anti-diabetes agents vs usual care/placebo/no intervention in adults with T2DM and normal cognition

Study name (year) country of conduct	Study design	Study risk of bias (measurement/ tool)	Mean age: intervention/ control	Sample size (intervention/ control)	Intervention	Comparator	Outcome domains with available data (synthesis method/metric)	Specific outcomes measure	Time point of measurement
Intervention: Intensive glucose control; Outcome: Incident dementia (1)									
The ADVANCE Collaborative Group (2008) (2)	RCT	High Quality (NOS)	66 / 66	11 140 (5 571/5 569) Female: 42.6% / 42.3%	Intensive glucose control	Standard glucose control	Dementia (RR)	First date of medication use, hospital records, or significant global cognitive decline	Median 5 years
20 countries									
Intervention: Intensive glucose control; Outcome: Cognitive function (1)									
Koekkoek et al 2012 (3) Netherlands	RCT	High Quality (NOS)	59.3 / 59.5	183 (97 / 86) Female: 42.3% / 35.7%	Intensive glucose control	Standard glucose control	Cognitive Function	Neuropsychological battery	10 years
Murray et al 2017(4) United States	RCT (Observational Ancillary)	High Quality (NOS)	62.1	1 328 (684 / 644) Female:	Intensive glucose control	Standard glucose Control	Cognitive Function	MMSE	80 months

of America			43.10%						
Luchsinger et al 2011(5) United States of America	RCT	High Quality (NOS)	70.6 / 70.4	2 169 (1 093 / 1076) Female: 61.8% / 60.2%	Intensive glucose control	Standard glucose control	Cognitive Function	CARE-Diagnostic Scale	3.5 years
Zimering et al 2016(6) United States of America	RCT	High Quality (NOS)	60.4	1 764 (877 / 887) Female: 4%	Intensive glucose control	Standard glucose control	Cognitive Function	Neuropsychological battery	5.6 years
Intervention: Intranasal insulin; Outcome: Cognitive function (7)									
Hallschmid et al 2008(8) Germany	RCT	Low (Cochrane RoB)	33.47 / 33.87	30 (15 / 15) Female: 0% / 0%	Insulin Actrapid	Placebo	Cognitive Function	Neuropsychological battery (Immediate and Delayed Word List Recall, Wordstem Priming Task, Stroop Test)	8 weeks
Novak et al 2014(9) United States of America	RCT	Low (Cochrane RoB)	62.0 / 60.1	29 (15 / 14)	Novolin R	Healthy Controls	Cognitive Function	Neuropsychological battery	Acute
Novak et al 2022(10) United States of America	RCT	Unclear (not reported)	63.6 / 65.7	106 (51 / 55) at baseline	Novolin R	Placebo	Cognitive Function	Neuropsychological battery (Cambridge Cognition Battery)	Approx 40 weeks.
Intervention: DPP-4i; Outcome: Cognitive function (11)									
Biessels et al 2019(12) Netherlands	RCT	Unclear (Cochrane RoB)	67.8 / 67.7	1 545 (800 / 745) Female: 37.1% /30.5%	Linagliptin	Placebo	Cognitive Function	MMSE	30 months
Biessels et al	RCT	Unclear	64.4 / 64.4	3 163 (1618 /	Linagliptin	Glimepiride	Cognitive	MMSE	40 months

2021(13) Netherlands		(Cochrane RoB)		1545) Female: 38.1% / 38.0%			Function		
Xue et al 2020(14) China	RCT	High (Cochrane RoB)	68.5 / 67.4	60 (30 / 30) Female: 43.3% /53.3%	DPP-4i (general)	Sulfonylurea	Cognitive Function	MMSE	6 months
Intervention: GLP1-RA; Outcome: Cognitive function (15)									
Cukierman- Yaffe et al 2020(16) 24 Countries	RCT	Unclear (CRoB)	65.5 / 65.5	8 826(4 456 / 4 372)	Dulaglutide	Placebo	Cognitive Function	DSST; MoCA	Median 5.4 years

Table WA.E.4b GRADE Summary of findings- Glucose-lowering medicines vs usual care/placebo/no intervention in adults with T2DM and MCI

Study name (year) country of conduct	Study design	Study risk of bias (measurement tool)	Mean age: intervention / control	Sample size (intervention/ control)	Intervention	Comparator	Outcome domains with available data (synthesis method/metric)	Specific outcomes measure	Time point of measurement
Intervention: DPP-4i; Outcome: Cognitive Function (17)									
Borzi et al 2019(18) Italy	RCT	Unclear (CRoB)	77.67	60 (30 / 30) Female: 55%	Vildagliptin	Metformin	Cognitive Function	MMSE	180 days
Ates et al 2019(19)	RCT	Unclear (CRoB)	74.4	95 (43 / 52) Female: 50%	Vildagliptin	Placebo	Cognitive Function	MMSE	180 days
Biessels et al 2021(13) Netherlands	RCT	Unclear (CRoB)	64.4	3 163 (1 618 / 1 545) Female: 38.0%	Linagliptin	Sulfonylureas	Cognitive Function	MMSE	160 weeks
Biessels et al	RCT	Low (CRoB)	67.8	1 545 (800 /	Linagliptin	Placebo	Cognitive Function	MMSE	2.5 years

2019(12) Netherlands				745) Female: 35.0%						
Guo et al 2021(20) China	RCT	Unclear (CRoB)	66.2	87 (43 / 44) Female: 49.4%	Sitagliptin	Sulfonylureas	Cognitive Function	MMSE	48 weeks	
Hu et al 2018(21) China	RCT	Unclear (CRoB)	66.8	98 (50 / 48) Female: 48.0	Linagliptin	Metformin	Cognitive Function	MMSE	24 weeks	
Jin et al 2017(22) China	RCT	High (CRoB)	72.57	80 (20 / 20) Female: 48.8% (general)	DPP-4i	Sulfonylureas	Cognitive Function	NNSE	180 days	
Xue et al 2019(14) China	RCT	Unclear (CRoB)	68.5	60 (30 / 30) Female: 48.3% (general)	DPP-4i	Sulfonylureas	Cognitive Function	MMSE	180 days	
Rizzo et al 2014(23) Italy	RCT	High (CRoB)	73.1	240 (120 / 120) Female: 60.8% (general)	DPP-4i	Metformin	Cognitive Function	MMSE	2 years	

Table WA.E.4c GRADE Summary of findings- Lifestyle/diet interventions vs usual care/placebo/no intervention in adults with T2DM and normal cognition

Study name (year) country of conduct	Study design	Study risk of bias (measurement tool)	Mean age: intervention /control	Sample size (intervention/ control)	Intervention	Comparator	Outcome domains with available data (synthesis method/metric)	Specific outcomes measure	Time point of measurement
Intervention: Intensive Diet and Exercise; Outcome: Incident Dementia or Incident MCI (24)									
Espeland et al 2017(25) United States of America	RCT	High (CRoB)	Range: 45–76	3 802 (1 918 / 1 884)	Intensive lifestyle (aiming to reach daily calorie goals of 1 200–1 800; 175m +/-week physical activity)	Diabetes support and education groups	Incident Dementia (Narrative); Incident MCI	Expert adjudication, based on 3MSE and Functional Assessment Questionnaire	Mean 11.4 years
Intervention: Exercise; Outcome: Cognitive Function (26)									
Chen et al 2023(27) China	RCT	Not reported at study level. Overall unclear risk of bias.	67.5	218 (107 / 111)	Tai chi chuan training	Workshops	Cognitive Function	MoCA	24 weeks
Ploydang et al 2023 United States of America (28)	RCT		68.9	33 (17 / 17)	Nordic aquatic walking	Maintaining daily activities	Cognitive Function	MoCA, MMSE	12 weeks
Kaligal et al 2023(29) India	RCT		52.5	50 (25 / 25)	Yoga	Maintaining daily activities	Cognitive Function	MoCA	12 weeks
Furlano et al 2023(30) Canada	RCT		68.2	24 (13 / 11)	Resistance training	Balance, stretch, and mobility exercises	Cognitive Function	Stroop Test	24 weeks

Ghodrati et al 2022(31) Canada	RCT	58.8	21 (12 / 9)	Aerobic, Resistance, and balance training	Maintaining daily activities	Cognitive Function	MoCA	12 weeks
Martinez-Velilla et al 2011(32) United States of America	RCT	87	103 (54 / 49)	Individualised resistance, balance, and walking.	Usual care	Cognitive Function	MMSE	-
Sotomayor et al 2021(33) Spain	RCT	Range: 64 – 78	76 (38 / 38)	Walking	Maintaining daily activities	Cognitive Function	MMSE	16 weeks
Yamamoto et al 2021(34) Japan	RCT	73.2	53 (18 / 17)	Resistance training (bodyweight)	Maintaining daily activities	Cognitive Function	MMSE	48 weeks
Sotomayor et al 2020(35) United States of America	RCT	72.3	107 (55 / 52)	Walking	Maintaining daily activities	Cognitive Function	MMSE	24 weeks
Zhu et al 2018 (36) China	RCT	70.3	60 (29 / 31)	Dance (Aerobic)	Maintaining daily activities	Cognitive Function	MoCA	12 weeks
Espeland et al 2017 United States of America (25)	RCT	Range: 70 - 89	415 (199 / 216)	Walking, strength, flexibility, and balance training	Workshops, stretching and flexibility exercises	Cognitive Function	3MSE	96 weeks

Callisaya et al 2017(37) Australia	RCT		65.3	50 (26 / 24)	Aerobic training and progressive resistance training	Stretching and gentle movement	Cognitive Function	Stroop Test	24 weeks
Intervention: Exercise; Outcome: Quality of Life (Mental and Physical) (38)									
MacDonald et al 2021(39) Denmark	RCT	High (CRoB)	53.6 / 56.5	98 (64 / 34) Female: 48% / 47%	Aerobic and resistance training	Standard care (medical counselling, lifestyle advice, education)	Quality of Life	SF-36	12 months
Myers et al 2013 (40) United States of America	RCT	High (CRoB)	56.9 / 58.1	77 (49 / 28) Female: 28% / 19%	Aerobic, Resistance, and combined training	Sham exercise (stretching and relaxation classes)	Quality of Life	SF-36	9 months
Nicolucci et al 2012(41) Italy	RCT	High (CRoB)	N/A. More than 50% of sample > 60.	538 (278 / 260)	Aerobic, resistance, and combined training	Standard Care	Quality of Life	SF-36	12 months
Reid et al 2010(42) Canada	RCT	High (CRoB)	53.3 / 55.2	109 (57 / 52) Female: 33.3% / 37%	Aerobic and resistance training	Nil (waiting list)	Quality of Life	SF-36	6 months
Stubbs Jr et al 2019(43) United States of America	RCT	High (CRoB)	63.0 / 61.0	16 (7 / 6)	Aerobic and resistance training	Standard care	Quality of Life	SF-36	12 weeks

Table WA.E.4d GRADE Summary of findings- Lifestyle/diet interventions vs usual care/placebo/no intervention in adults with T2DM and MCI

Study name (year) country of conduct	Study design	Study risk of bias (measurement tool)	Mean age: intervention control	Sample size /(intervention/ control)	Intervention	Comparator	Outcome domains with available data (synthesis method/metric)	Specific outcomes measure	Time point of measurement
Intervention: DPP-4i; Outcome: Cognitive Function (17)									
Wang et al 2023 (44) China	RCT	Not reported at study level. Overall unclear risk of bias.	65.73 / 68.07	82 (40 / 42)	Aerobic exercise and relaxation training	Routine Care	Cognitive Function	MoCA, MMSE	12 months
Chen et al 2023 (27) China	RCT		67.56 / 67.62	218 (107 / 111)	Tai chi	Maintaining daily activities	Cognitive Function	MoCA, MMSE	24–36 weeks
Wang et al 2015(45) China	RCT		62.37 / 64.51	89 (43 / 46)	Baduanjin, Relaxation Gong	Routine Care	Cognitive Function	MoCA, MMSE	12 months
Yamamoto et al 2021(34) Japan	RCT		73.2 / 73.3	53 (36 / 17)	Resistance training	Routine Care	Cognitive Function	MMSE	48 weeks
Zhang et al 2017(46) China	RCT		69.53 / 69.79	74 (36 / 38)	Aerobic training & walking	Routine Care	Cognitive Function	MoCA	3 months
Zhu et al 2015 (47) China	RCT		69.92	78 (37 / 41)	Baduanjin	Routine Care	Cognitive Function	MoCA	12 months

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